

SGA 6 – Expose V, continued

Generalities on λ -rings; the Chern ring and Adams operations

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2. Pre- λ -rings, continued

2.3.4

For every $x \in K$ and every integer $n \geq 1$ one has

$$(-1)^{n-1} n \lambda^n(x) = \sum_{i=1}^n (-1)^{i-1} \lambda^{n-i}(x) \psi^i(x). \quad (1)$$

This formula makes it possible to express the operations λ^n in terms of the operations ψ^n , and conversely.

2.3.5

Put

$$\lambda_{-1/t}(x)(-t)^n = \sum_{k=0}^n (-1)^k \lambda^k(x) t^{n-k}. \quad (2)$$

Then

$$\lambda^n(\lambda^m(x)) = P_{n,m}(\lambda^1(x), \dots, \lambda^{nm}(x)), \quad (3)$$

where $P_{n,m}$ is a universal isobaric polynomial of weight nm .

2.3.6

Let K be a pre- λ -ring. We define a filtration $\text{Fil}^\bullet K$ of K by putting

$$\text{Fil}^0 K = K, \quad \text{Fil}^1 K = K, \quad \text{Fil}^n K = \text{the subgroup generated by the } x \text{ such that } \lambda^i(x) = 0 \text{ for } i < n. \quad (4)$$

One checks that

$$\text{Fil}^p K \cdot \text{Fil}^q K \subset \text{Fil}^{p+q} K.$$

Grothendieck operations

2.3.7

Let K be a pre- λ -ring. The *Grothendieck operations* γ^i are defined by

$$\gamma_t(x) = \lambda_{t/(1-t)}(x) = \sum_{n \geq 0} \gamma^n(x) t^n. \quad (5)$$

Equivalently,

$$\gamma^n(x) = \lambda^n(x + n - 1) = \sum_{i=0}^n (-1)^i \binom{n-1}{i} \lambda^{n-i}(x). \quad (6)$$

2.3.8

The following formulas hold:

$$\gamma^0(x) = 1, \quad \gamma^1(x) = x, \quad \gamma^n(x + y) = \sum_{i+j=n} \gamma^i(x) \gamma^j(y). \quad (7)$$

The operations γ^i define a new pre- λ -ring structure, called the *opposite* structure.

2.3.9

One has

$$\gamma^n(xy) = P'_n(\gamma^1(x), \dots, \gamma^n(x); \gamma^1(y), \dots, \gamma^n(y)), \quad (8)$$

where P'_n is a universal polynomial.

2.4

A λ -ring is a pre- λ -ring K such that the homomorphism

$$\lambda_t : K \longrightarrow 1 + K[[t]]_+$$

is a λ -homomorphism, that is, is compatible with the pre- λ -ring structures. This means that

$$\lambda^n(\lambda^m(x)) = P_{n,m}(\lambda^1(x), \dots, \lambda^{nm}(x)) \quad (9)$$

for all x and all n, m , where $P_{n,m}$ is the universal polynomial of 2.3.5.

2.4.1

The following conditions are equivalent:

- (i) K is a λ -ring.
- (ii) The Adams operations satisfy $\psi^n \circ \psi^m = \psi^{nm}$ for all n, m .
- (iii) For every finite-dimensional element x , the elements $\lambda^n(x)$ satisfy the universal relations.

2.4.2

If K is a λ -ring, the Adams operations ψ^n are ring endomorphisms and satisfy

$$\psi^n \circ \psi^m = \psi^{nm}. \quad (10)$$

If K is also augmented, the operations ψ^n are compatible with the augmentation.

2.5

Let K be a pre- λ -ring. Denote by $\mathfrak{W}(K)$ the multiplicative group $1 + K[[t]]_+$. We have seen that $\mathfrak{W}(K)$ carries a pre- λ -ring structure. If K is a λ -ring, then $\mathfrak{W}(K)$ is again a λ -ring.

2.5.1

The element $1 + t$ is the unit for the product $\circ$. One has

$$\lambda^n(1 + t) = 1 \quad (n \geq 2), \quad \psi^n(1 + t) = 1 + t^n. \quad (11)$$

2.6

Let K be a ring. We denote by $\mathfrak{W}_\circ(K)$ the functor $\mathfrak{W}$ equipped with the product $\circ$ defined in 2.3 and with its operations λ^i . We denote by $\mathfrak{W}_\gamma(K)$ the same functor equipped with the product and operations arising from the operations γ^i .

2.6.1

There is a commutative diagram

$$\begin{array}{ccc} \mathfrak{W}_\circ(K) & \xrightarrow{\lambda_t} & \mathfrak{W}_\circ(\mathfrak{W}_\circ(K)) \\ \downarrow & & \downarrow \\ \mathfrak{W}_\gamma(K) & \xrightarrow{\gamma_t} & \mathfrak{W}_\gamma(\mathfrak{W}_\gamma(K)). \end{array}$$

3. The γ -operations

3.1

Let K be a ring and let $\mathfrak{W}(K) = 1 + K[[t]]_+$. The opposite pre- λ -ring structure on K defines operations γ^i , related to the operations λ^i by

$$\gamma_t(x) = \lambda_{t/(1-t)}(x). \quad (12)$$

3.2

For every $x \in K$ and every integer $n \geq 0$,

$$\gamma^n(x) = \lambda^n(x + n - 1). \quad (13)$$

This follows from the definition and the binomial formula for exterior powers.

3.3

Let K be a pre- λ -ring. We say that K is a γ -ring if the operations γ^i define a λ -ring structure on K , that is, if

$$\gamma_t : K \longrightarrow \mathfrak{W}_\gamma(K)$$

is a λ -homomorphism.

3.4

The following conditions are equivalent:

- (i) K is a γ -ring.
- (ii) The operations γ^i satisfy the universal relations.
- (iii) The operations ψ^n satisfy $\psi^n \circ \psi^m = \psi^{nm}$.

3.5

It is immediate that a λ -ring is a γ -ring and conversely; indeed the conditions expressed in terms of the ψ^n are identical. Moreover,

$$\gamma^n(xy) = \sum_{i+j=n} P'_{i,j}(\gamma^1(x), \dots; \gamma^1(y), \dots), \quad (14)$$

where the $P'_{i,j}$ are the universal polynomials of 2.3.9.

3.6

Let K be a ring. The functor $\mathfrak{W}_\gamma(K)$ is a γ -ring, just as $\mathfrak{W}_\circ(K)$ is a λ -ring. There is a canonical isomorphism

$$\mathfrak{W}_\circ(K) \simeq \mathfrak{W}_\gamma(K), \quad (15)$$

given by the change of variable $t \mapsto t/(1-t)$.

3.7

Let K be a λ -ring and let A be a K - λ -algebra. The canonical γ -filtration of A is defined by

$$\text{Fil}_\gamma^n A = \text{exttheidealgeneratedbytheproducts} \gamma^{i_1}(x_1) \cdots \gamma^{i_k}(x_k) \quad (16)$$

with $\sum i_j \geq n$ and $\varepsilon(x_j) = 0$. It is decreasing:

$$A = \text{Fil}_\gamma^0 A \supset \text{Fil}_\gamma^1 A \supset \cdots \supset \text{Fil}_\gamma^n A \supset \cdots,$$

and

$$\text{Fil}_\gamma^0 A = A, \quad \text{Fil}_\gamma^1 A = \text{Ker}(\varepsilon), \quad \text{Fil}^p A \cdot \text{Fil}^q A \subset \text{Fil}^{p+q} A. \quad (17)$$

3.8

Let K be a binomial ring and let A be a K - λ -algebra. Then

$$\gamma_t(x) = (1/(1-t))^x = \exp(-x \log(1-t)). \quad (18)$$

If $a \in K$, multiplication by $\gamma_t(a)$ for the γ -law is characterized by its value on $1 + Xt$:

$$\gamma_t(a)(1 + Xt) = (1 + Xt)^a = \exp(a \log(1 + Xt)) = \sum_{n=0}^{\infty} \binom{a}{n} X^n t^n. \quad (19)$$

In particular the n -th universal polynomial defining multiplication by $\gamma_t(a)$ for the γ -law is the polynomial $R_{n,a}$ of 2.9.

3.9–3.10

Let A be an augmented K - λ -algebra, with augmentation $\varepsilon : A \rightarrow K$. The *canonical filtration* of A is the ring filtration for which $\text{Fil}^n A$ is the ideal generated by products

$$\gamma^{i_1}(x_1) \cdots \gamma^{i_k}(x_k), \quad \sum i_j \geq n, \quad \varepsilon(x_j) = 0. \quad (20)$$

Thus

$$\text{Fil}^0 A = A, \quad \text{Fil}^1 A = \text{Ker } \varepsilon. \quad (21)$$

This filtration is functorial in A : every λ -homomorphism of augmented K - λ -algebras is compatible with the filtrations. Unless expressly stated otherwise, this is the filtration considered on an augmented λ -ring.

3.11

Let A be a λ -ring. Denote by $\text{Fil}_\lambda^n A$ the filtration defined using the operations λ^i , equivalently the canonical γ -filtration of the opposite structure. One has

$$\text{Fil}_\lambda^n A = \text{Fil}_\gamma^n A. \quad (22)$$

Indeed the two filtrations are defined by the same conditions, in view of the equivalence between λ -rings and γ -rings.

3.11.1–3.11.2

Let A be a λ -ring and let $x_1, \dots, x_n$ be elements of A with $\varepsilon(x_i) = 0$. Then

$$\gamma^n(x_1 + \cdots + x_n) \equiv x_1 \cdots x_n \pmod{\text{Fil}^{n+1} A}. \quad (23)$$

If $x_1, \dots, x_n$ are elements of filtration at least 1, then

$$\gamma^n(x_1 + \cdots + x_n) \equiv \sum_{i=1}^n x_i \pmod{\text{Fil}^2 A}. \quad (24)$$

4. λ -rings generated by generators and relations

4.1

We work throughout over $\mathcal{C}_0$. Let K be a λ -ring. Consider a family of indeterminates $(X_i)_{i \in I}$ and put

$$A = K[(X_i), (\lambda^n X_i)_{i \in I, n \geq 2}].$$

We wish to define on A a λ -ring structure extending that of K . To give a homomorphism $\lambda_t : A \rightarrow \mathfrak{W}(A)$, it is enough to give the images of the X_i and of the $\lambda^n X_i$; these determine a ring homomorphism $A \rightarrow \mathfrak{W}(A)_\circ$. We put

$$\lambda_t(X_i) = 1 + X_i t + \sum_{n \geq 2} \lambda^n X_i t^n, \quad \lambda_t(\lambda^n X_i) = \lambda^n(\lambda_t(X_i)). \quad (25)$$

Since the relevant diagram is a diagram of ring homomorphisms, it is enough to check commutativity on the X_i and $\lambda^n X_i$; this follows from the definitions and from the fact that λ_u and $\mathfrak{W}^u(\lambda_t)$ are λ -homomorphisms.

4.2

The λ -ring A is characterized by the following universal property.

Proposition. *For every K - λ -algebra B and every family $(b_i)_{i \in I}$ of elements of B , there exists a unique K - λ -homomorphism $f : A \rightarrow B$ such that $f(X_i) = b_i$ for all $i \in I$.*

For f to be a λ -homomorphism it is necessary that $f(\lambda^j X_i) = \lambda^j(b_i)$ for all i, j , and this determines f uniquely. The resulting homomorphism is a λ -homomorphism because the equality $\lambda_t \circ f = \mathfrak{W}(f) \circ \lambda_t$ need only be checked on generators of A over K .

4.3–4.4

One may make the same construction using the γ 's instead of the λ 's. Let

$$A' = K[(X_i), (\gamma^n X_i)_{i \in I, n \geq 2}].$$

The structure is defined by

$$\gamma_t(X_i) = 1 + X_i t + \sum_{n \geq 2} \gamma^n X_i t^n, \quad \gamma_t(\gamma^n X_i) = \gamma'^n(\gamma_t(X_i)). \quad (26)$$

The same universal property as in 4.2 holds, hence A' is canonically isomorphic, as a λ -ring, to A .

The ring A is called the *free λ -ring* generated over K by the family $(X_i)_{i \in I}$.

Let A be a λ -ring and J an ideal of A . We say that J is a λ -ideal if for every $x \in J$ and every $n \geq 1$ one has $\lambda^n(x) \in J$. The kernel of a λ -homomorphism is a λ -ideal. Conversely, if J is a λ -ideal, then A/J inherits a λ -ring structure.

4.5–4.7

An intersection of λ -ideals is a λ -ideal. Hence there is a smallest λ -ideal containing any given family $(x_i)_{i \in I}$; by the relations of 2.4.2 it is the ordinary ideal generated by the $\lambda^n(x_i)$, $n \geq 1$.

Let $(P_{i'})_{i' \in I'}$ be a family of elements in the free λ -ring $A = K[(X_i), (\lambda^n X_i)]$. If J is the λ -ideal generated by the $P_{i'}$, then A/J is called the λ -ring generated over K by the generators $(X_i)_{i \in I}$ subject to the relations $(P_{i'})_{i' \in I'}$. The same construction can be made with relations among the X_i and the $\gamma^n X_i$, since an ideal is a λ -ideal if and only if it is stable under all operations γ^n .

If B is a K - λ -algebra and (b_i) is a family of elements of B , the relations $P_{i'}$ can be evaluated at $b_i, \lambda^n(b_i)$. If all these evaluations vanish, then there is a unique K - λ -homomorphism $A/J \rightarrow B$ sending X_i to b_i .

4.8–4.9

Let $n \geq 0$. In the free λ -ring on one indeterminate X over K , the ideal generated by the $\lambda^k X$, $k > n$, is a λ -ideal. Indeed, if $A = K[X, \lambda^k X]_{k \geq 2}$ and $B = A/J$, one has to show that the composite $A \xrightarrow{\lambda^t} \mathfrak{W}(A) \rightarrow \mathfrak{W}(B)$ vanishes on J . This reduces to the following elementary fact: for any ring C and any series $\varphi \in \mathfrak{W}(C)$ whose coefficients vanish in degrees $> n$, one has $\lambda^k(\varphi) = 1$ for $k > n$. It is enough to prove this for

$$C = K[X_1, \dots, X_n], \quad \varphi = 1 + X_1 t + \dots + X_n t^n.$$

After adjoining roots by the elementary symmetric functions, one may write

$$1 + X_1 t + \dots + X_n t^n = \prod_{i=1}^n (1 + T_i t),$$

and therefore

$$\lambda_u(1 + X_1 t + \dots + X_n t^n) = \prod_{i=1}^n \lambda_u(1 + T_i t) = \prod_{i=1}^n (1 + \{1 + T_i t\}u),$$

which has degree n in u .

Corollary. *The λ -ring generated over K by one generator X subject to the relations $\lambda^k(X) = 0$ for $k > n$ is isomorphic to $K[X, \dots, \lambda^n X]$.*

The same assertions hold with λ replaced by γ . One may also replace X by a family (X_i) subject to relations $\lambda^k(X_i) = 0$ for $k > n_i$.

4.10–4.15

Let K be a binomial ring and A an augmented K - λ -algebra. Let

$$B = A[X, \gamma^k X]_{k \geq 2}$$

be the free λ -ring on X over A , augmented by sending X to 0. We grade B by the weight in the variables $\gamma^k X$, giving $\gamma^k X$ weight k , and denote by M_k the set of monomials of weight k .

Lemma. *Let $a \in A$. The coefficient of a monomial of weight n in*

$$\gamma^k(aX) = P'_k(\gamma^i(a), \gamma^j X)$$

has filtration at least $k - n$ in A .

The polynomial P'_k has no isobaric component of weight $< k$, so the lemma is clear when a has augmentation zero. In general one writes

$$\gamma_t(aX) = \gamma_t((a - \varepsilon(a))X) \gamma_t(\varepsilon(a)X),$$

and uses

$$\gamma_t(\varepsilon(a)X) = (\gamma_t X)^{\varepsilon(a)} = \sum_{n=0}^{\infty} R_{n, \varepsilon(a)}(\gamma^j X) t^n,$$

with $R_{n, a}$ isobaric of weight n .

Proposition. *One has*

$$\mathrm{Fil}^k B = \mathrm{Fil}^k A + \mathrm{Fil}^{k-1} A \cdot M_1 + \cdots + \mathrm{Fil}^0 A \cdot M_k + B \cdot M_{k+1}. \quad (27)$$

The inclusion of the right hand side in the left is clear. Conversely, since the right hand side defines a ring filtration, it suffices to treat the generators $\gamma^k(b)$ with $\varepsilon(b) = 0$. Writing $b = \sum a_i m_i$ with m_i a monomial in the $\gamma^j X$, additivity reduces the assertion to $\gamma^k(am_i)$. Using the lemma and the A - λ -homomorphism sending X to m_i , one reduces further to $\gamma^k(m_i)$. Finally the absence of components of weight $< k$ in the polynomials P'_k and $Q'_{j, k}$ gives the result.

The decomposition in (27) is a direct sum decomposition of A -modules: if

$$b = \sum_{j \leq k} a_{i, j} m_{i, j} + b'$$

with $m_{i, j} \in M_j$ and $b' \in B \cdot M_{k+1}$, then $b \in \mathrm{Fil}^k B$ implies $a_{i, j} \in \mathrm{Fil}^{k-j} A$. The assertion also remains valid after quotienting by any λ -ideal contained in the augmentation ideal, and with any family of indeterminates instead of the single X .

Corollary.

$$\mathrm{Gr}^k B \simeq \mathrm{Gr}^k A \oplus \mathrm{Gr}^{k-1} A \cdot M_1 \oplus \cdots \oplus \mathrm{Gr}^0 A \cdot M_k.$$

If B' is the λ -ring generated over K by X with relations $\gamma^k(X) = 0$ for $k > n$, so $B' = K[X, \dots, \gamma^n X]$, then its λ -ring filtration is the same as the weight filtration in the variables $\gamma^j X$.

5. The elements $\lambda^p(N, x)$

This section will only be used in Expose VII, section 4, for the study of regular immersions; it may be omitted on a first reading.

5.1–5.3

Let $d \geq 1$. Let A be the $\mathbb{Z}$ - λ -algebra generated by N and x , subject to the relations $\lambda^i(N) = 0$ for $i > d$. Thus

$$A = \mathbb{Z}[N, \dots, \lambda^d N; x, \dots, \lambda^j x, \dots].$$

Put

$$\lambda_{-1}(N) = \sum_{i=0}^d (-1)^i \lambda^i(N).$$

Lemma. *For every $p \geq 1$, $\lambda^p(x\lambda_{-1}(N))$ is divisible by $\lambda_{-1}(N)$.*

Since $\lambda^p(x\lambda_{-1}(N))$ is isobaric of weight p in the quantities $\lambda^j(\lambda_{-1}(N))$, it suffices to prove the lemma for $\lambda^p(\lambda_{-1}(N))$. Let B be the λ -ring generated over $\mathbb{Z}$ by $N_1, \dots, N_d$ with $\lambda^i(N_j) = 0$ for $i \geq 2$. Sending N to $\sum_i N_i$ defines a λ -homomorphism $A \rightarrow B$ which identifies A with the symmetric polynomials in the N_i . Then

$$\lambda_{-1}(N) = \prod_{i=1}^d (1 - N_i).$$

If $N_i = 1$ for some i , then $\lambda_t(\lambda_{-1}(N))$ becomes $\lambda_t(0) = 1$, so all $\lambda^p(\lambda_{-1}(N))$ vanish. Hence each is divisible by every $1 - N_i$, and therefore by their product.

Since A is integral, we define $\lambda^p(N, x)$ unambiguously by

$$\lambda^p(N, x)\lambda_{-1}(N) = \lambda^p(x\lambda_{-1}(N)), \quad p \geq 1. \quad (28)$$

Thus $\lambda^p(N, x)$ is a polynomial in the $\lambda^i(x)$ and $\lambda^j(N)$, isobaric of weight p in the $\lambda^i(x)$. If a λ -ring B contains an element N'' with $\lambda^k(N'') = 0$ for $k > d$, then $\lambda^p(N'', x'')$ is defined by substitution; it does not depend on the chosen bound d .

5.4

In the λ -ring generated by N, x, y subject to $\lambda^k(N) = 0$ for $k > d$, formula (28) gives

$$\begin{aligned} \lambda^p(N, x+y)\lambda_{-1}(N) &= \lambda^p((x+y)\lambda_{-1}(N)) \\ &= \sum_{i+j=p} \lambda^i(x\lambda_{-1}(N))\lambda^j(y\lambda_{-1}(N)). \end{aligned}$$

Since the ring is integral, this yields the identity

$$\lambda^p(N, x+y) = \sum_{\substack{i+j=p \\ i,j \neq 0}} \lambda^i(N, x)\lambda^j(N, y)\lambda_{-1}(N) + \lambda^p(N, x) + \lambda^p(N, y). \quad (29)$$

If N' is another element with $\lambda^k(N') = 0$ for $k > d'$, then $\lambda^k(N + N') = 0$ for $k > d + d'$, and additivity gives

$$\lambda^p(N + N', x)\lambda_{-1}(N) = \lambda^p(N', x\lambda_{-1}(N)). \quad (30)$$

5.5

Let A be any λ -ring and let $N \in A$ satisfy $\lambda^k(N) = 0$ for $k > d$. We associate to A a new λ -ring A_N . First define on A a new multiplication by

$$x \cdot_N y = xy\lambda_{-1}(N). \quad (31)$$

This ring generally has no unit, so we adjoin a unit and consider $\mathbb{Z} \times A$, with

$$(n, x) + (n', x') = (n + n', x + x')$$

and

$$(n, x)(n', x') = (nn', nx' + n'x + x \cdot_N x'). \quad (32)$$

The ring A embeds by $x \mapsto (0, x)$. Define a pre- λ -ring structure by

$$\lambda_t(n, x) = (1 + t)^n \sum_{p=0}^{\infty} \lambda^p(N, x) t^p. \quad (33)$$

It is a λ -ring: by substitution it suffices to treat the universal case in which A is generated over $\mathbb{Z}$ by N, X, Y with $\lambda^k(N) = 0$ for $k > d$. There is then a homomorphism $A_N \rightarrow A$ sending $(1, 0)$ to 1 and $(0, x)$ to $x\lambda_{-1}(N)$; by (31) and (28) it is a λ -homomorphism, and it is injective in the universal integral case. Since A is a λ -ring, so is A_N .

In A_N one can form the γ -operations. For $x \in A$,

$$\gamma^p(0, x) = \sum_{i=0}^{p-1} \binom{p-1}{i} \lambda^i(N, x),$$

which lies in A ; equivalently it is characterized by

$$\gamma^p(N, x)\lambda_{-1}(N) = \gamma^p(x\lambda_{-1}(N)). \quad (34)$$

5.6

Let A be the λ -ring generated over $\mathbb{Z}$ by N, L, x subject to $\lambda^k(N) = 0$ for $k > d$ and $\lambda^k(L) = 0$ for $k > 1$. We compute $\lambda^p(N + L, x)$ in terms of the $\lambda^q(N, x)$. By definition,

$$\lambda^p(N + L, x)\lambda_{-1}(N)\lambda_{-1}(L) = \lambda^p(x\lambda_{-1}(N)\lambda_{-1}(L)),$$

hence

$$\begin{aligned} 1 + \lambda_{-1}(N)\lambda_{-1}(L) \sum_{k \geq 1} \lambda^k(N + L, x)t^k &= \lambda_t(x\lambda_{-1}(N)) \circ \lambda_t(1 - L) \\ &= \frac{\lambda_t(x\lambda_{-1}(N))}{\lambda_{Lt}(x\lambda_{-1}(N))} \\ &= 1 + \frac{\sum_{k \geq 1} \lambda^k(N, x)\lambda_{-1}(N)(1 - L^k)t^k}{1 + \sum_{k \geq 1} \lambda^k(N, x)\lambda_{-1}(N)L^k t^k}. \end{aligned}$$

5.7

Let K be a binomial ring, A an augmented K - λ -algebra, and $N \in A$ such that $\lambda^k(N) = 0$ for $k > d$. Then for every $x \in A$ and every $p \geq 1$,

$$\gamma^p(N, x) \in \text{Fil}^{p-d} A. \quad (35)$$

Indeed, let $A_\circ$ be the K - λ -algebra generated by $N'_\circ, X'_\circ$ with $\lambda^k(N'_\circ) = 0$ for $k > d$, and augmented by $\varepsilon(N'_\circ) = \varepsilon(X'_\circ) = 0$. This is the polynomial ring over K in the variables $\gamma^i(N'_\circ)$, $1 \leq i \leq d$, and $\gamma^j(X'_\circ)$, $j \geq 1$; its λ -filtration is the weight filtration. Put

$$N_\circ = N'_\circ + d, \quad X_\circ = X'_\circ + \varepsilon(x).$$

For $N_\circ, X_\circ$ the assertion is equivalent to

$$\gamma^p(N_\circ, X_\circ)\gamma^d(N'_\circ) \in \text{Fil}^p A_\circ.$$

Since $\gamma^d(N'_\circ) = (-1)^d \lambda_{-1}(N_\circ)$, this becomes

$$(-1)^d \gamma^p(X_\circ \gamma^d(N'_\circ)) \in \text{Fil}^p A_\circ,$$

which holds because $\gamma^d(N'_\circ)$ has zero augmentation. A K - λ -homomorphism $A_\circ \rightarrow A$ compatible with augmentations sends $N'_\circ$ to $N - d$ and $X'_\circ$ to $x - \varepsilon(x)$; compatibility with filtrations gives the desired result.

End of Expose V

SGA 6 – Expose V, continued

The Chern ring and Adams operations

6. Chern ring

Throughout this section the base ring is a fixed binomial ring K . We shall use the category $\mathcal{G}$ of 1.1, and therefore return to the notation of section 1: $\mathfrak{G}$ denotes the functor defined on $\mathcal{G}$, and $\mathfrak{G}_0$ its restriction to $\mathcal{G}_0$.

6.1

In 3.5 and 3.6 we defined morphisms of functors

$$\mathfrak{G}_0 \times \mathfrak{G}_0 \xrightarrow{*} \mathfrak{G}_0, \quad \mathfrak{G}_t \circ i \xrightarrow{\gamma'} \mathfrak{G}_u \circ i \circ \mathfrak{G}_{t\underline{s}}, \quad \mathfrak{G}_0 \xrightarrow{a} \mathfrak{G}_0,$$

the last being multiplication, for the law $\circ$, by $\gamma^t(a)$ with $a \in K$. By 1.9 these yield morphisms

$$\mathfrak{G} \times \mathfrak{G} \xrightarrow{*} \mathfrak{G}_g, \quad \mathfrak{G}_t \xrightarrow{\gamma'} \mathfrak{G}_u \circ i \circ \mathfrak{G}_{t\underline{s}}, \quad \mathfrak{G} \xrightarrow{a} \mathfrak{G}_g.$$

The isobaricity of the polynomials $P'_n, Q'_{i,j}, R_{n,a}$ implies that the images of the first and third functors lie in $\mathfrak{G}$, and that the image of the second lies in the subgroup of formal series in u whose non-zero degree coefficients lie in $\mathfrak{G}_t$.

Composing the first and third morphisms with the canonical morphism $\mathfrak{G} \rightarrow 1 + \mathfrak{G}^+$ gives two morphisms of functors on $\mathcal{G}$:

$$(1 + \mathfrak{G}^+) \times (1 + \mathfrak{G}^+) \xrightarrow{*} 1 + \mathfrak{G}^+, \quad 1 + \mathfrak{G}^+ \xrightarrow{a} 1 + \mathfrak{G}^+.$$

They define on $1 + \mathfrak{G}^+$ a multiplication and operations of K . The multiplication, denoted $*$, is determined by its value on $(1 + X, 1 + Y)$:

$$(1 + X) * (1 + Y) = \frac{1 + X + Y}{(1 + X)(1 + Y)}. \quad (36)$$

The n -th universal polynomial for this law is the isobaric component of weight n of P'_n , denoted P''_n . The operations of K are again written exponentially; the universal polynomials defining them are the lowest-degree isobaric components of the $R_{n,a}$, and since the $R_{n,a}$ are themselves isobaric, these components are the $R_{n,a}$.

For $A \in \text{Ob}(\mathcal{G})$, the ring $1 + \widehat{A}^+$ defined by these laws generally has no unit. We extend it by adjoining the scalars of K , obtaining the ring $K \times (1 + \widehat{A}^+)$ with

$$(a, \varphi) + (a', \varphi') = (a + a', \varphi\varphi')$$

and

$$(a, \varphi)(a', \varphi') = (aa', \varphi^{a'} \varphi'^a (\varphi * \varphi')). \quad (37)$$

The unit is $(1, 1)$, and $1 + \widehat{A}^+$ is the ideal $(0) \times (1 + \widehat{A}^+)$. It is naturally a K -algebra, with

$$a \cdot (a', \varphi) = (a, 1)(a', \varphi) = (aa', \varphi^a). \quad (38)$$

The preceding construction of γ' and the canonical morphism $\mathfrak{G} \rightarrow 1 + \mathfrak{G}^+$ give a morphism

$$\mathfrak{G}_t \xrightarrow{\gamma'} \mathfrak{G}_u \circ i \circ (K \times (1 + \mathfrak{G}^+)),$$

and therefore a morphism of functors

$$1 + \mathfrak{G}^+ \longrightarrow \mathfrak{G}_u \circ i \circ (K \times (1 + \mathfrak{G}^+)).$$

The universal polynomials defining it are the coefficients of $\gamma'^i(1 + X_1 + \cdots + X_n + \cdots)$; the j -th coefficient is the isobaric component of weight j of $Q'_{i,j}$, denoted $Q''_{i,j}$.

Extending this morphism to $K \times (1 + \mathfrak{G}^+)$ using the λ -structure of K , one obtains a pre- λ -ring structure whose corresponding γ -operations satisfy

$$\gamma_t(a, \varphi) = \gamma_t(a) \left(1 + \sum_{p \geq 1} \gamma'^p(\varphi) t^p \right). \quad (39)$$

We write these operations simply as γ . The verification that this is a λ -ring structure reduces to the commutativity of the diagram of 3.7.2; it is known on K and follows on $1 + \mathfrak{G}^+$ from the corresponding diagram for $\mathfrak{G}$ by passing to the quotient.

Definition. *The functor defined above on graded K -algebras, with values in augmented K - λ -algebras, is called the Chern ring. Its value on A is denoted $\text{Ch}(A)$, or $\text{Ch}_K(A)$ if the base must be specified.*

From (36), (37), (39), and 3.6.2, one obtains

$$(1, 1 + X)(1, 1 + Y) = (1, 1 + X + Y), \quad \lambda^i(1, 1 + X) = 0 \quad (i \geq 2). \quad (40)$$

These are the relations used to define the Chern ring in Grothendieck's treatment. Conversely, they determine (36) and (39).

This construction is motivated as follows. Suppose that A is the associated graded ring of the $K^\bullet$ of a commutatively locally ringed topos, for the filtration of 3.10, and that this $K^\bullet$ is a λ -ring. We shall define Chern classes of elements of an augmented λ -ring with values in its associated graded ring. To a locally free sheaf of finite type one then associates an element of $\text{Ch}(A)$: its scalar component is the rank, and its $1 + \widehat{A}^+$ component is the sum of its Chern classes. The formulas (40) then become the usual formulas for tensor products of invertible sheaves and exterior powers of an invertible sheaf.

6.3. The Chern character

For every graded K -algebra A , denote by $A_{\mathbb{Q}}$ the graded K -algebra $A \otimes_{\mathbb{Z}} \mathbb{Q}$. Define a functorial homomorphism

$$1 + \widehat{A}^+ \xrightarrow{\text{ch}} \widehat{A}_{\mathbb{Q}}^+$$

by its value on $1 + X$:

$$\text{ch}(1 + X) = \exp(X) - 1. \quad (41)$$

This homomorphism is compatible with the multiplication of 6.1. Indeed, it is enough to check this on $1 + X, 1 + Y$, and one computes

$$\begin{aligned} \text{ch}((1 + X) * (1 + Y)) &= \text{ch}\left(\frac{1 + X + Y}{(1 + X)(1 + Y)}\right) \\ &= \text{ch}(1 + X + Y) - \text{ch}(1 + X) - \text{ch}(1 + Y) \\ &= \exp(X + Y) - \exp(X) - \exp(Y) + 1 \\ &= \text{ch}(1 + X) \text{ch}(1 + Y). \end{aligned}$$

Let η be the functorial endomorphism of $\widehat{A}_{\mathbb{Q}}^+$ defined by

$$\eta\left(\sum_{k \geq 1} a_k t^k\right) = \sum_{k \geq 1} \frac{(-1)^{k-1}}{(k-1)!} a_k t^k.$$

Then, for $\varphi \in 1 + \widehat{A}^+$,

$$\text{ch}(\varphi) = \eta(\text{Log } \varphi). \quad (42)$$

It follows that ch commutes with the operations of K , since this may again be checked on $1 + X$:

$$\text{ch}((1 + X)^a) = \eta(a \text{Log}(1 + X)) = a\eta(\text{Log}(1 + X)) = a \text{ch}(1 + X).$$

Thus ch extends to a homomorphism of K -algebras

$$\text{ch} : K \times (1 + \widehat{A}^+) \longrightarrow K \oplus \widehat{A}_{\mathbb{Q}}^+, \quad \text{ch}(a, \varphi) = a + \eta(\text{Log } \varphi). \quad (43)$$

This homomorphism is called the *Chern character*. If A has characteristic zero, the Chern character is an isomorphism, with inverse

$$a + \psi_1 \longmapsto (a, \exp(\eta^{-1}\psi_1)).$$

Proposition. *Let A be a graded K -algebra. Give $\text{Ch}(A)$ the degree filtration*

$$(\text{Ch } A)_0 = \text{Ch } A, \quad (\text{Ch } A)_n = 0 \times (1 + \widehat{A}^{\geq n}) \quad (n \geq 1), \quad (44)$$

where $\widehat{A}^{\geq n} = \prod_{i \geq n} A^i$. Then:

(i) *If $\varphi \in (\text{Ch } A)_n$, $n \geq 1$, and $\varphi = 1 + a_n + \dots$, then*

$$\text{ch}(\varphi) = \frac{(-1)^{n-1}}{(n-1)!} a_n + \dots. \quad (45)$$

(ii) *If $\varphi = 1 + a_m + \dots \in (\text{Ch } A)_m$ and $\psi = 1 + b_n + \dots \in (\text{Ch } A)_n$, then*

$$\varphi * \psi = 1 - \frac{(m+n-1)!}{(m-1)!(n-1)!} a_m b_n + \dots. \quad (46)$$

In particular the degree filtration is a ring filtration.

The first congruence follows immediately from (42). For the second, by substitution one reduces to a torsion-free universal polynomial ring, embeds it in its tensor product with $\mathbb{Q}$, and uses the injectivity of ch . If $\varphi * \psi = 1 + Z_p + \dots$ is the first non-zero term, multiplicativity of ch and (45) identify Z_p .

6.6. Comparison of filtrations

The λ -ring $\text{Ch } A$ is augmented over K by $(a, \varphi) \mapsto a$. Hence it has its canonical filtration as an augmented λ -ring. We compare it with the degree filtration.

Proposition. *Let $x \in \text{Ch } A$ have the form $x = (n, 1 + \dots + a_n)$. Then:*

- (i) $\lambda^i(x) = 0$ for $i > n$;
- (ii) $\lambda^n(x) = (1, 1 + a_1)$;
- (iii) $\sum_{i=0}^n (-1)^i \lambda^i(x) = (0, 1 - (n-1)!a_n + \dots)$, where the omitted terms have degree $> n$.

It is enough, by functoriality, to prove this for $A = K[X_1, \dots, X_n]$ filtered by weight, with X_i of weight i , and $x = (n, 1 + X_1 + \dots + X_n)$. Embedding by symmetric functions into $K[T_1, \dots, T_n]$ gives

$$(n, 1 + X_1 + \dots + X_n) = \sum_{i=1}^n (1, 1 + T_i).$$

Then (40) gives the assertions on $\lambda^k(x)$, and the product formula (46) gives the final congruence for $\lambda_{-1}(x)$.

Corollary. *The relation 6.6.1(iii) holds for every element x of augmentation n .*

For this one uses that each $(\text{Ch } A)_n$ is a λ -ideal. If $x = (0, \varphi) \in (\text{Ch } A)_n$, then the polynomials defining the coefficients of $\lambda^i(x)$ depend only on the first coefficients of φ ; hence all coefficients of degree $< n$ vanish. Thus $\lambda^i(x) \in (\text{Ch } A)_n$.

Consequently, for $x = (n, 1 + a_1 + \dots + a_n + \dots)$,

$$\lambda_{-1}(x) = (0, 1 - (n-1)!a_n + \dots).$$

Equivalently, if ε is the augmentation of $\text{Ch } A$, then

$$\lambda^n(x - \varepsilon(x)) = (0, 1 + (-1)^{n-1}(n-1)!a_n + \dots). \quad (47)$$

It follows that the canonical λ -ring filtration satisfies

$$\text{Fil}^n \text{Ch } A \subset (\text{Ch } A)_n. \quad (48)$$

If A is a finite graded K -algebra of characteristic zero, the two filtrations are in fact identical. Indeed, if $A^{\geq p} = 0$ and $x = (0, 1 + a_n + \dots + a_p) \in (\text{Ch } A)_n$, put

$$x' = \left(0, 1 + \frac{(-1)^{n-1}}{(n-1)!} a_n + a_{n+1} + \dots + a_p \right).$$

Then $\lambda^n(x') = (0, 1 + a_n + \dots)$, and one proceeds by induction on n .

6.7. Chern classes in an augmented λ -ring

Let A be an augmented λ -ring over K , with augmentation ε . Since A has its canonical filtration, one may form the Chern ring of $\text{Gr } A$. For $x \in A$, the element $\gamma_t(x - \varepsilon(x))$ lies in $\mathfrak{G}(A)$; its image in

$$\mathfrak{G}(A)/\mathfrak{G}'(A) = 1 + \widehat{\text{Gr } A}^+$$

is denoted $c(x)$ and is called the *total Chern class* of x . Its i -th coefficient is denoted $c^i(x)$:

$$c^i(x) = \text{cl}_{\text{Gr}^i A}(\lambda^i(x - \varepsilon(x))), \quad c(x) = 1 + c^1(x) + \cdots. \quad (49)$$

Put

$$\tilde{c}(x) = (\varepsilon(x), c(x)), \quad (50)$$

called the *completed Chern class* of x .

Proposition. *The completed Chern class*

$$\tilde{c} : A \longrightarrow \text{Ch}(\text{Gr } A)$$

is a homomorphism of augmented K - λ -algebras.

Additivity is clear. For multiplicativity, write

$$\begin{aligned} \gamma_t(xy - \varepsilon(xy)) &= \gamma_t(\varepsilon(y)(x - \varepsilon(x)))\gamma_t(\varepsilon(x)(y - \varepsilon(y))) \\ &\quad \cdot \gamma_t((x - \varepsilon(x))(y - \varepsilon(y))) \\ &= (\gamma_t(x - \varepsilon(x)))^{\varepsilon(y)}(\gamma_t(y - \varepsilon(y)))^{\varepsilon(x)} \\ &\quad \cdot (\gamma_t(x - \varepsilon(x))\overline{\circ}\gamma_t(y - \varepsilon(y))), \end{aligned}$$

and pass to the associated graded, using (37). Compatibility with scalars and augmentations is immediate. Compatibility with the λ -operations follows by checking the diagram

$$\begin{array}{ccc} A & \xrightarrow{\gamma_t} & \mathfrak{G}_g(A) \\ \downarrow \tilde{c} & & \downarrow \mathfrak{G}_g(\tilde{c}) \\ \text{Ch}(\text{Gr } A) & \xrightarrow{\gamma_t} & \mathfrak{G}_g(\text{Ch}(\text{Gr } A)) \end{array}$$

on K and on elements of augmentation zero.

Proposition. *Let A be an augmented K - λ -algebra and let $z \in \text{Fil}^m A$. Then*

$$\gamma^m(z) \equiv (-1)^{m-1}(m-1)!z \pmod{\text{Fil}^{m+1} A}. \quad (51)$$

Equivalently,

$$c^m(z) = \text{cl}_{\text{Gr}^m A}((-1)^{m-1}(m-1)!z).$$

One writes z as a sum of products $a\gamma^{i_1}(x_1)\cdots\gamma^{i_k}(x_k)$ with $\varepsilon(x_i) = 0$ and $i_1 + \cdots + i_k \geq m$. Since $\tilde{c}$ is a λ -homomorphism and (47) controls the first non-zero term of the Chern class, formula (46) gives the displayed congruence.

Proposition. *Let K be a binomial ring, R an augmented K - λ -algebra, $N \in R$ such that $\lambda^k(N) = 0$ for*

$k > d$, and $x \in \text{Fil}^i R$. Then

$$\gamma^{d+i}(N, x) - (-1)^{d+i-1}(d+i-1)! x \in \text{Fil}^{i+1} R. \quad (52)$$

Choose a presentation of x by products of γ -operations, and pass to the universal K - λ -algebra R_0 generated by N'_0 and the auxiliary variables, with $\gamma^k(N'_0) = 0$ for $k > d$ and zero augmentation. The filtration of R_0 is the weight filtration. After replacing N'_0 by $N_0 + d$, the desired assertion becomes

$$\gamma^{d+i}(N_0, x_0) \gamma^d(N'_0) - (-1)^{d+i-1}(d+i-1)! x_0 \gamma^d(N'_0) \in \text{Fil}^{d+i+1} R_0.$$

Since $\lambda_{-1}(N_0) = (-1)^d \gamma^d(N'_0)$, this is exactly Proposition 6.9 applied to $z = (-1)^d x_0 \gamma^d(N'_0)$. Functoriality then gives the general case.

Proposition. *Assume that K has characteristic zero. Let $n \geq 0$. Then the functor Ch is an equivalence from the category of graded K -algebras A with $A_0 = K$ and $A_k = 0$ for $k > n$ to the category of augmented K - λ -algebras Λ with $\text{Fil}^{n+1} \Lambda = 0$. A quasi-inverse is $\Lambda \mapsto \text{Gr } \Lambda$.*

If A is finite graded, the two filtrations on $\text{Ch } A$ coincide. Define

$$u : A \longrightarrow \text{Gr}_{\deg}(\text{Ch } A)$$

by the identity on A_0 and, for $i > 0$, by

$$u(a_i) = \text{cl}_{\text{Gr}^i}(0, 1 + (-1)^{i-1}(i-1)! a_i).$$

Formula (46) shows that u is a homomorphism of graded rings, compatible with the K -operations. Its inverse sends a class in $\text{Gr}^i(\text{Ch } A)$ to $(-1)^{i-1} a_i / (i-1)!$, where a_i is the i -th coefficient of any lift.

For an augmented λ -algebra Λ , the isomorphism

$$\Lambda \longrightarrow \text{Ch}(\text{Gr } \Lambda)$$

is given by the completed Chern class $\tilde{c}$. It is enough to check this on associated graded rings, and then on generators of the canonical filtration, i.e. the classes $\gamma^i(x)$ with $\varepsilon(x) = 0$. Formula (47) shows that the resulting composite is the identity.

Corollary. *If K has characteristic zero and Λ is a K - λ -algebra with discrete filtration, then*

$$\tilde{c} : \Lambda \longrightarrow \text{Ch}(\text{Gr } \Lambda)$$

is an isomorphism.

Definition. *Let A be a graded K -algebra and Λ an augmented K - λ -algebra. A Chern theory for Λ with values in A is a λ -homomorphism*

$$\Lambda \longrightarrow \text{Ch}(A)$$

compatible with augmentations.

The functor sending A to the set of such Chern theories is representable. When K has characteristic zero and Λ has finite filtration, the representing graded algebra is $\text{Gr } \Lambda$ by Proposition 6.11.

7. Appendix: Adams operations ψ^k

We follow here a presentation due to J.-P. Serre.

7.1

Let K be a pre- λ -ring. For $x \in K$ define elements $\psi^k(x)$ by

$$\frac{d}{dt} \lambda_t(x) / \lambda_t(x) = \sum_{k=1}^{\infty} (-1)^{k-1} \psi^k(x) t^{k-1}. \quad (53)$$

The maps $\psi^k : K \rightarrow K$ are additive, since logarithmic derivation turns products into sums. They are isobaric polynomials of weight k in $\lambda^1, \dots, \lambda^k$; for example

$$\psi^1(x) = x, \quad \psi^2(x) = x^2 - 2\lambda^2(x).$$

7.2–7.3

Conversely, if K is a ring of characteristic zero and additive maps $\psi^k : K \rightarrow K$ are given with $\psi^1 = \text{id}_K$, then

$$\lambda_t(x) = \exp \left(\sum_{k=1}^{\infty} (-1)^{k-1} \psi^k(x) t^k / k \right) \quad (54)$$

defines a pre- λ -structure on K . Thus in characteristic zero giving a pre- λ -structure is equivalent to giving the corresponding operations ψ^k . It is also clear from (53) that the ψ^k commute with λ -homomorphisms; in characteristic zero a homomorphism of pre- λ -rings is a λ -homomorphism if and only if it commutes with all ψ^k .

7.4

For $K = \mathbb{Z}$ with its λ -ring structure, all ψ^k are the identity. Indeed

$$\frac{d}{dt} \lambda_t(n) / \lambda_t(n) = n(1+t)^{n-1} / (1+t)^n = n \sum_{k \geq 1} (-1)^{k-1} t^{k-1}.$$

On the category $\mathcal{G}_0$ of $\mathbb{Z}$ -algebras, the operations ψ^k define additive endomorphisms of the λ -ring functor $\mathfrak{G}$. They are determined by

$$\psi^k(1 + Xt) = 1 + X^k t. \quad (55)$$

Proposition. (i) *If K is a λ -ring, the operations ψ^k satisfy*

$$\psi^k(1) = 1, \quad \psi^k(xy) = \psi^k(x)\psi^k(y), \quad \psi^k \circ \psi^{k'} = \psi^{kk'}. \quad (56)$$

(ii) *Conversely, if the ψ^k satisfy these identities and K has characteristic zero, then K is a λ -ring.*

For the first assertion, $\lambda_t(1) = 1 + t$ gives $\psi^k(1) = 1$. To prove multiplicativity, consider the diagram expressing the product $K \times K \rightarrow K$ and the product $\circ$ on $\mathfrak{G}(K)$; after applying logarithmic derivative and

extracting the $k - 1$ coefficient, it is enough to check the universal pair $(1 + Xt, 1 + Yt)$. Then

$$\psi^k((1 + Xt) \circ (1 + Yt)) = \psi^k(1 + XYt) = (XY)^k,$$

which equals $X^k Y^k$.

For the composition law it is enough to check the universal element again:

$$\psi_u^{k'}(\lambda_u(1 + Xt)) = \psi_u^{k'}(1 + (1 + Xt)u) = (1 + Xt)^{\circ k'} = 1 + X^{k'}t,$$

and applying ψ^k gives $X^{kk'}$. If K has characteristic zero, logarithmic derivation is an isomorphism from $\mathfrak{G}(K)$ to $K[[t]]$; hence the relations in (56) force λ_t to be a ring homomorphism and then a λ -homomorphism. The relation $\lambda_t(1) = 1 + t$ follows by integration from $\psi^k(1) = 1$.

Bibliography for Expose V

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SGA 6 – Expose VI

$K^\bullet$ of a Projective Bundle: Calculations and Consequences

Pierre Berthelot

1. Computation of K^* of a projective bundle: locally free sheaves of finite type

In this section $K^*(X)$ denotes the Grothendieck group of locally free sheaves of finite type on a scheme X . The aim is the following theorem.

Theorem (1.1). *Let S be a quasi-compact scheme, let E be a locally free $\mathcal{O}_S$ -module of rank $r + 1$, and let $X = \mathbb{P}(E)$ be the associated projective bundle. The structural morphism $f : X \rightarrow S$ gives $K^*(X)$ the structure of a $K^*(S)$ -algebra. Let $\xi \in K^*(X)$ be the class of $\mathcal{O}_X(1)$. Then $K^*(X)$ is a free $K^*(S)$ -module with basis*

$$1, \xi, \dots, \xi^r.$$

1.2. Plan of the proof

The proof has three parts.

1°) The powers of ξ generate $K^*(X)$ as a $K^*(S)$ -module. This is proved by showing:

- (a) If $\mathcal{F}$ is locally free of finite type on X , there is a graded $\mathrm{Sym}(E)$ -module $\mathcal{M}$ of finite presentation, flat over $\mathcal{O}_S$, such that $\mathcal{F} = \mathrm{Proj}_0(\mathcal{M})$.
- (b) Such an $\mathcal{M}$ has a finite graded resolution by graded $\mathrm{Sym}(E)$ -modules $\mathcal{L}_i$ locally free of finite type:

$$0 \rightarrow \mathcal{L}_{r+1} \rightarrow \dots \rightarrow \mathcal{L}_0 \rightarrow \mathcal{M} \rightarrow 0.$$

- (c) Each $\mathcal{L}_i$ has a finite filtration whose quotients are of the form $\mathcal{N}_j \otimes_{\mathcal{O}_S} \mathrm{Sym}(E)(j)$ with $\mathcal{N}_j$ locally free of finite type on S .

2°) The elements $1, \xi, \dots, \xi^{r+1}$ are linearly dependent over $K^*(S)$.

3°) The elements $1, \xi, \dots, \xi^r$ are linearly independent over $K^*(S)$.

1.3. Graded modules

Let S be a scheme, $\mathcal{B}$ a graded $\mathcal{O}_S$ -algebra, and $\mathcal{M}$ a graded $\mathcal{B}$ -module. For a graded $\mathcal{B}$ -module $\mathcal{N}$, write $\mathcal{N}(n)$ for the graded module with

$$(\mathcal{N}(n))_k = \mathcal{N}_{n+k}.$$

A graded module is locally free, respectively locally free of finite type, if locally on S it is a direct sum, respectively a finite direct sum, of modules $\mathcal{B}(n)$. It is of finite presentation if locally it has a presentation by finite free graded modules.

For every n there is a canonical homomorphism of graded $\mathcal{B}$ -modules

$$\mathcal{M}_n \otimes_{\mathcal{O}_S} \mathcal{B}(-n) \longrightarrow \mathcal{M}, \tag{57}$$

sending $x \otimes a$ to $a \cdot x$. A graded module free of finite type decomposes as

$$\mathcal{M} \simeq \bigoplus_{n=n_0}^{n_1} (\mathcal{B}(-n))^{k_n}. \quad (58)$$

The integers k_n in such a decomposition are independent of the chosen decomposition. This is proved by induction, comparing the lowest non-zero homogeneous components and then passing to the quotient.

Proposition. *Let S be quasi-compact, E a locally free $\mathcal{O}_S$ -module of finite type, $X = \mathbb{P}(E)$, and $\mathcal{F}$ an $\mathcal{O}_X$ -module of finite presentation, flat over S . There exists an integer n_0 such that for all $n \geq n_0$:*

- (i) $R^i f_*(\mathcal{F}(n)) = 0$ for $i \geq 1$;
- (ii) $f_*(\mathcal{F}(n))$ is locally free of finite type on S ;
- (iii) $\mathcal{M} = \bigoplus_{n \geq n_0} f_*(\mathcal{F}(n))$ is a graded $\mathrm{Sym}(E)$ -module of finite presentation.

The assertion is local on S , hence the quasi-compactness allows reduction to S affine. In the noetherian case, (i) follows from Serre's theorem, (ii) from EGA III, 7.9.10, and (iii) from the comparison between $\mathcal{F}$ and a quasi-coherent graded module defining it. For the general affine case, write S as a filtered projective limit of noetherian affine schemes and descend E and $\mathcal{F}$ to some stage. The necessary flatness and finite-presentation hypotheses descend by EGA IV, and the noetherian case applies after base change.

Corollary. *Under the hypotheses of Proposition VI.1.3, there is a graded $\mathrm{Sym}(E)$ -module $\mathcal{M}$ of finite presentation, flat over $\mathcal{O}_S$, such that $\mathcal{F} = \mathrm{Proj}_0(\mathcal{M})$.*

Lemma. *Let A be a ring, B an A -algebra of finite presentation, and M a B -module of finite presentation. Assume that B and M are flat over A , and that for every residue field k of A , the module $M \otimes_A k$ has projective dimension $\leq n$ over $B \otimes_A k$. Then M has a resolution of length n by projective B -modules of finite type:*

$$0 \rightarrow L_n \rightarrow \cdots \rightarrow L_0 \rightarrow M \rightarrow 0.$$

This follows immediately from EGA IV, 12.3.2.

Proposition. *Let S be quasi-compact, E locally free of rank $r+1$, $\mathcal{B} = \mathrm{Sym}(E)$ with its usual grading, and $\mathcal{M}$ a graded $\mathcal{B}$ -module of finite presentation, flat over S . Then there is a graded resolution*

$$0 \rightarrow \mathcal{L}_{r+1} \rightarrow \cdots \rightarrow \mathcal{L}_0 \rightarrow \mathcal{M} \rightarrow 0,$$

with the $\mathcal{L}_i$ graded $\mathcal{B}$ -modules locally free of finite type.

First, any such $\mathcal{M}$ admits a graded surjection $\mathcal{L} \rightarrow \mathcal{M}$ with $\mathcal{L}$ locally free of finite type: choose N such that $\mathcal{M}$ is generated in degrees $\leq N$ and put

$$\mathcal{L} = \bigoplus_{i \leq N} \mathcal{M}_i \otimes_{\mathcal{O}_S} \mathcal{B}(-i).$$

The map is the one of (57). Iterating this construction on kernels gives a global graded resolution. After $r+1$ steps, the kernel is locally projective over $\mathcal{B}$ on affine opens by Lemma VI.1.5, hence locally free as a graded $\mathcal{B}$ -module; truncating gives the asserted resolution.

Proposition. *Let S be quasi-compact, E locally free of finite type, $\mathcal{B} = \text{Sym}(E)$, and $\mathcal{L}$ a graded $\mathcal{B}$ -module locally free of finite type. Then $\mathcal{L}$ has a decreasing finite filtration by graded submodules $\mathcal{L}^{(m)}$ such that*

- (i) $\mathcal{L}^{(m)} = \mathcal{L}$ for $m \leq m_0$;
- (ii) $\mathcal{L}^{(m)} = 0$ for $m \geq m_1$;
- (iii) $\text{gr}^m \mathcal{L}$ is isomorphic, as a graded $\mathcal{B}$ -module, to $\mathcal{N}_m \otimes_{\mathcal{O}_S} \mathcal{B}(m)$ with $\mathcal{N}_m$ locally free of finite type on S .

Since S is quasi-compact, the degrees of local homogeneous bases lie between fixed integers n_0 and n_1 . Induction on $n_1 - n_0$ proves the assertion. The first step is the canonical map $\mathcal{M}_{n_0} \otimes \mathcal{B}(-n_0) \rightarrow \mathcal{M}$, which locally identifies the summand of lowest homogeneous degree; its cokernel has local generators in higher degrees, and induction applies.

1.8–1.9. Generation and the relation

Let $\mathcal{F}$ be locally free of finite type on X . By Corollary VI.1.4 and Proposition VI.1.6, resolve the corresponding graded module by $\mathcal{L}_i$'s. Applying Proj_0 gives an exact sequence and hence in $K^*(X)$

$$\text{cl}(\mathcal{F}) = \sum_i (-1)^i \text{cl}(\text{Proj}_0 \mathcal{L}_i).$$

Using the filtration of Proposition VI.1.7, each term is a sum of classes

$$\text{cl}(f^* \mathcal{N}_{ij}(j)) = f^*(\text{cl} \mathcal{N}_{ij}) \xi^j.$$

Thus $K^*(X)$ is generated as a $K^*(S)$ -module by the powers of ξ .

To get a dependence relation, use the canonical surjection $f^* E \rightarrow \mathcal{O}_X(1)$ with locally free kernel $\mathcal{F}$ of rank r . Since $\lambda^{r+1}(f^* E - \xi) = 0$, and $\lambda_t(-\xi) = 1/(1 + \xi t)$, one obtains

$$\sum_{i=0}^{r+1} (-1)^{r+1-i} \xi^{r+1-i} \lambda^i(E) = 0. \quad (59)$$

1.10–1.11. Independence

The proof of independence is due to J.-P. Jouanolou.

Lemma. *With the hypotheses of Theorem VI.1.1, let $\mathcal{F}$ and $\mathcal{G}$ be locally free sheaves of finite type on X with equal classes in $K^*(X)$. Then there is n_0 such that for all $n \geq n_0$:*

- (i) $f_*(\mathcal{F}(n))$ and $f_*(\mathcal{G}(n))$ are locally free of finite type on S ;
- (ii) their classes in $K^*(S)$ are equal.

This follows by writing the equality of classes in $K^*(X)$ as a finite integral combination of relations coming from short exact sequences, twisting by $\mathcal{O}_X(n)$, and applying f_* . For n sufficiently large all higher direct images vanish and all direct images are locally free, so the relations descend to $K^*(S)$.

Suppose

$$f^*(x_0) + f^*(x_1)\xi + \cdots + f^*(x_r)\xi^r = 0. \quad (60)$$

Write each x_j as a difference of classes of locally free sheaves, and construct locally free sheaves $\mathcal{F}, \mathcal{F}'$ on X such that (60) says $\text{cl}(\mathcal{F}) = \text{cl}(\mathcal{F}')$. By the lemma, for large n ,

$$\sum_{j=0}^r x_j \sigma_{n+j} = 0, \quad (61)$$

where σ_p is the class of $\text{Sym}^p(E)$ in $K^*(S)$.

The Koszul complex

$$0 \rightarrow \lambda^{r+1}(E) \otimes \text{Sym}(E) \rightarrow \lambda^r(E) \otimes \text{Sym}(E) \rightarrow \cdots \rightarrow E \otimes \text{Sym}(E) \rightarrow \text{Sym}(E) \rightarrow 0$$

is a graded resolution of $\mathcal{O}_S$. Its homogeneous components give

$$\sum_{i=0}^{r+1} (-1)^i \lambda^i(E) \sigma_{p-i} = 0 \quad (p \geq 1), \quad (62)$$

with $\sigma_k = 0$ for $k < 0$. From (61) for all large n , relation (62) allows descending induction on n , and gives (61) for all $n \geq -r$. Taking $n = -r$ gives $x_r = 0$, and continuing downward gives $x_j = 0$ for all j . This proves independence and Theorem VI.1.1.

Remark. Let $\sigma_i(E)$ denote the class of $\text{Sym}^i(E)$. If $\sigma_t(E) = 1 + \sum_{i \geq 1} \sigma_i(E) t^i$, relation (62) is equivalent to

$$\lambda_t(E) \sigma_{-t}(E) = 1. \quad (63)$$

This shows the additivity of σ_t and extends it to a homomorphism $K^*(S) \rightarrow 1 + K^*(S)[[t]]^+$.

Remark. Any sequence of $r+1$ consecutive powers of ξ is a basis of $K^*(X)$ over $K^*(S)$. Put $x = 1 - \xi^{-1}$, the class of a hyperplane section. Then $1, x, \dots, x^r$ is also a basis, and the dependence relation is

$$\sum_{k=0}^{r+1} \lambda^{r+1-k}(\check{E} - r - 1) x^k = 0. \quad (64)$$

For $E = \mathcal{O}_S^{r+1}$, so $X = \mathbb{P}_S^r$, this reduces to

$$x^{r+1} = 0. \quad (65)$$

Remark. Using relative schemes in the sense of Hakim, Theorem VI.1.1 also makes sense when S is a locally ringed topos. If the final object of the topos is quasi-compact, the proof transports without difficulty.

2. Computation of K^* of a projective bundle: perfect complexes

This section will not be used later in the seminar. Here $K^*(X)$ denotes the Grothendieck group of perfect complexes on X .

Theorem (2.1). Let S be quasi-compact, let E be a locally free $\mathcal{O}_S$ -module of rank $r+1$, and let $X = \mathbb{P}(E)$. Let ξ be the class in $K^*(X)$ of the complex consisting of $\mathcal{O}_X(1)$ in degree zero. Then $K^*(X)$ is a free $K^*(S)$ -module with basis

$$1, \xi, \dots, \xi^r.$$

Let $\mathcal{B} = \text{Sym}(E)$. Let $\mathcal{A}$ be the category of $\mathcal{O}_X$ -modules, $\mathcal{G}$ the category of graded $\mathcal{B}$ -modules, $\mathcal{G}_0$ its full

subcategory of graded $\mathcal{B}$ -modules locally free of finite type, and $\mathcal{G}'$ the category of $\mathcal{O}_S$ -modules. For a complex L^* on X , write $L^*(n) = L^* \otimes \mathcal{O}_X(n)$.

2.3. The functor $\underline{\text{Proj}}_0$

We need to apply Proj_0 to arbitrary graded $\mathcal{B}$ -modules, not only quasi-coherent ones. If $Y = \text{Spec}(\mathcal{A})$ and $f : Y \rightarrow S$, the isomorphism $\mathcal{A} \simeq f_* \mathcal{O}_Y$ gives a morphism of ringed spaces

$$\tilde{f} : (Y, \mathcal{O}_Y) \longrightarrow (S, \mathcal{A})$$

and an exact inverse-image functor

$$\tilde{f}^*(\mathcal{M}) = f^{-1} \mathcal{M} \otimes_{f^{-1} \mathcal{A}} \mathcal{O}_Y.$$

For a quasi-coherent $\mathcal{A}$ -module this is the usual associated module.

Now let $\mathcal{B}$ be graded and quasi-coherent, $\mathcal{M}$ a graded $\mathcal{B}$ -module, and $X = \text{Proj}(\mathcal{B})$. For an affine open $U \subset S$ and a homogeneous section s of positive degree, put

$$\mathcal{M}_{(s),U} = ((\mathcal{M}|_U) \otimes_{\mathcal{B}|_U} (\mathcal{B}|_U)_s)_0.$$

On $D_+(U, s)$ define

$$\underline{\text{Proj}}_0(\mathcal{M})_{U,s} = \tilde{f}_{U,s}^*(\mathcal{M}_{(s),U}).$$

The usual localization isomorphisms

$$\mathcal{M}_{(st),U} \simeq (\mathcal{M}_{(s),U})_{t^d/s^e}$$

show that these modules glue. This gives an exact functor $\underline{\text{Proj}}_0$ on graded modules and hence a functor

$$\underline{\text{Proj}}_0 : D^+(\mathcal{G}) \longrightarrow D^+(\mathcal{A}),$$

compatible with tensor products.

2.4–2.6. Acyclic approximation

For an $\mathcal{O}_X$ -module $\mathcal{L}$ and an integer k , set

$$F_k(\mathcal{L}) = \bigoplus_{n \geq k} f_*(\mathcal{L}(n)). \quad (66)$$

The functor F_k is left exact, hence has a derived functor

$$\text{R}F_k : D^+(\mathcal{A}) \longrightarrow D^+(\mathcal{G}). \quad (67)$$

If $\mathcal{L}$ is locally free of finite type, then for k sufficiently large it is F_k -acyclic, by the vanishing theorem used in 1.3.

There is a functorial morphism

$$\beta : \underline{\text{Proj}}_0(F_k(\mathcal{F})) \longrightarrow \mathcal{F} \quad (68)$$

constructed on each standard open $D_+(U, s)$ by sending a fraction z/s^n to $z\alpha_d(s^n)^{-1}$. It agrees with the usual EGA morphism when $\mathcal{F}$ is quasi-coherent. Passing to derived categories gives

$$\underline{\mathrm{Proj}}_0 \circ \mathrm{R}F_k \longrightarrow \mathrm{id}_{D^+(\mathcal{A})}. \quad (69)$$

Proposition. *Let S be quasi-compact, E locally free of finite type, $X = \mathbb{P}(E)$, and L^* a perfect complex on X . For all sufficiently large k :*

(i) $\mathrm{R}F_k(L^*)$ is $\mathcal{G}_0$ -perfect;

(ii) the canonical morphism

$$\underline{\mathrm{Proj}}_0 \mathrm{R}F_k(L^*) \longrightarrow L^*$$

is an isomorphism in $D^+(\mathcal{A})$.

The assertion is local on S , so assume S affine. Since $\mathcal{O}_X(1)$ is ample, one may replace L^* by a strictly perfect complex. For large k all its finitely many terms are F_k -acyclic and the modules $F_k(L^i)$ are $\mathcal{G}_0$ -perfect, giving (i); the morphism is then an isomorphism term by term, giving (ii).

2.7–2.8. Decomposition and proof

For each n , the functor taking the n -th homogeneous component gives

$$\mathrm{Comp}_n : D^+(\mathcal{G}) \longrightarrow D^+(\mathcal{G}').$$

The canonical map (57) gives a morphism

$$\mathrm{Comp}_n \otimes_{\mathcal{O}_S} \mathcal{B}(-n) \longrightarrow \mathrm{id}_{D^+(\mathcal{G})}. \quad (70)$$

If M^* is $\mathcal{G}_0$ -perfect, one constructs a finite sequence of distinguished triangles

$$N_i^* \rightarrow P_{i-1}^* \rightarrow P_i^* \rightarrow N_i^*[1]$$

with $P_0^* = M^*$, final P_k^* acyclic, and

$$N_i^* = K_i^* \otimes_{\mathcal{O}_S} \mathcal{B}(d_i)$$

for perfect complexes K_i^* on S . The construction starts with the smallest degree d_0 appearing locally in a bounded locally free representative, uses (70), and iterates. Finiteness follows from the finite filtration of Proposition VI.1.7.

Now let L^* be perfect on X . For large k , put $M^* = \mathrm{R}F_k(L^*)$. Applying the preceding decomposition and then $\underline{\mathrm{Proj}}_0$, one gets in $K^*(X)$

$$\mathrm{cl}(L^*) = \sum_i \mathrm{cl}(f^* K_i^*) \xi^{d_i} = \sum_i f^*(\mathrm{cl} K_i^*) \xi^{d_i}.$$

Thus the powers of ξ generate $K^*(X)$ over $K^*(S)$. The dependence relation follows from (59), viewed in the perfect-complex K -group. For independence one uses the $K^*(S)$ -linear map $f_* : K^*(X) \rightarrow K^*(S)$. Since

$$f_*(1) = 1, \quad f_*(\xi^n) = 0 \quad (-r \leq n < 0),$$

multiplying a relation by appropriate powers of ξ^{-1} and applying f_* successively gives all coefficients zero. This proves Theorem VI.2.1.

3. Consequences of the structure theorem

Proposition. *Let S be quasi-compact, let $E_1, \dots, E_n$ be locally free $\mathcal{O}_S$ -modules of finite type, and let X be the fiber product over S of the corresponding flag schemes. Then the canonical homomorphism*

$$K^*(S) \longrightarrow K^*(X)$$

is injective, whether K^ is defined using locally free sheaves of finite type or perfect complexes.*

By constructing flag schemes by successive projective bundles, this reduces to the projective-bundle case, where it follows immediately from Theorem VI.1.1 or VI.2.1.

Theorem. *For every quasi-compact scheme S , the pre- λ -ring $K^*(S)$ defined from locally free sheaves of finite type is a λ -ring.*

It is enough to verify the universal λ -relations on classes of locally free sheaves. After decomposing S into finitely many disjoint opens one may assume constant ranks. Pulling back to $X = \mathbb{P}(E) \times_S \mathbb{P}(F)$ is injective on K^* , and on X the relevant classes split into sums of invertible sheaves. Thus it suffices to check the relations for classes ξ, η of invertible sheaves, where they become

$$1 + \xi\eta t = (1 + \xi t) \circ (1 + \eta t), \quad \lambda_u(1 + \xi t) = 1 + \{1 + \xi t\}u,$$

which are precisely the defining relations of the universal λ -ring structure of $1 + K^*(S)[[t]]_+$.

More generally, if $(T, \mathcal{A})$ is a commutatively ringed topos and K is the Grothendieck group of locally free $\mathcal{A}$ -modules of bounded rank on T , then the canonical pre- λ -structure on K is a λ -ring. The proof is the same after reducing constant-rank locally free modules to representations of the group schemes $\mathrm{GL}(n)$ and using the fact that the representation ring $R_{\mathbb{Z}}(G)$ is a λ -ring for finite products of linear group schemes.

4. Computation of $K^\bullet$ of a flag bundle

4.1

Let S be a scheme and let $\mathcal{E}$ be a locally free $\mathcal{O}_S$ -module of rank n . Let $\pi = (p_1, \dots, p_k)$ be a sequence of integers ≥ 1 with $\sum_i p_i = n$. The *flag functor of type π of $\mathcal{E}$* assigns to an S -scheme $X \rightarrow S$ the set of chains of quotients of $f^*\mathcal{E}$

$$(\mathcal{L}_{k-1}, \dots, \mathcal{L}_1),$$

where $\mathcal{L}_i$ is a quotient of $\mathcal{L}_{i+1}$, locally free of rank $p_1 + \dots + p_i$. This functor is represented by an S -scheme $\underline{D}_\pi(\mathcal{E})$ equipped with the universal quotient flag. For $\pi = (1, n-1)$ this is $\mathbb{P}(\mathcal{E})$; for $\pi = (r, n-r)$ it is the Grassmannian; if all $p_i = 1$, it is the usual flag bundle $\underline{D}(\mathcal{E})$.

Lemma (4.2). *Let $\pi = (p_1, \dots, p_i, \dots, p_k)$ and split $p_i = p' + p''$. The flag bundle of type*

$$\pi' = (p_1, \dots, p_{i-1}, p', p'', p_{i+1}, \dots, p_k)$$

is obtained from $\underline{D}_\pi(\mathcal{E})$ by taking the flag bundle of type (p', p'') of the kernel of the relevant universal quotient. This identification is canonical.

The proof is immediate from the functorial definition of flag schemes.

Lemma (4.3). *Let A be a commutative ring, and let $t_1, \dots, t_m \in A$. Consider the algebra generated by variables $\xi_i^{(j)}$ subject to the elementary symmetric relations*

$$\sigma_r(\xi_1^{(j)}, \dots, \xi_m^{(j)}) = e_r(t_1, \dots, t_m).$$

For a partition $p_1 + \dots + p_k = n$, the corresponding block monomials

$$\prod_{j=1}^k \prod_{\alpha=1}^{p_j} (\xi_{p_1 + \dots + p_{j-1} + \alpha}^{(j)})^{a_{j,\alpha}}, \quad 0 \leq a_{j,\alpha} \leq n - \alpha,$$

form a basis over the subalgebra generated by the block elementary symmetric functions, and the relations among those symmetric functions are generated by the isobaric relations from

$$\prod_{j=1}^k \prod_{\alpha=1}^{p_j} (1 + \eta_j^{(\alpha)} T) = \prod_{i=1}^m (1 + t_i T).$$

Corollary (4.4). *The same block monomials form a basis over A .*

The lemma and corollary are borrowed from Chevalley's seminar.

4.5–4.8

Let S be quasi-compact, let $\mathcal{E}$ be locally free of rank n , and put $X = \underline{D}_\pi(\mathcal{E})$. Let $\mathcal{L}_i$ be the universal quotients and $\mathcal{F}_i = \text{Ker}(\mathcal{L}_i \rightarrow \mathcal{L}_{i-1})$. Denote by F_i the class of $\mathcal{F}_i$.

Theorem (4.6). *With this notation, the $K^\bullet(S)$ -algebra $K^\bullet(X)$ is generated by the Chern roots $\xi_i^{(\alpha)}$ of the F_i . The ideal of relations among them is generated by the relations deduced from*

$$\prod_{i=1}^k \lambda_t(F_i) = \lambda_t(E). \tag{71}$$

Equivalently, $K^\bullet(X)$ is the $K^\bullet(S)$ - λ -algebra generated by elements $F_i^{(\alpha)}$ subject to

- (i) $\lambda^r(F_i^{(\alpha)}) = 0$ for $r > p_i$;
- (ii) $\prod_{i,\alpha} (1 + F_i^{(\alpha)} t) = \lambda_t(E)$.

First consider the usual complete flag bundle. Induction on the rank, using the projective bundle theorem at each step, gives a $K^\bullet(S)$ -basis

$$\xi_1^{a_1} \xi_2^{a_2} \dots \xi_n^{a_n}, \quad 0 \leq a_i \leq n - i.$$

The relation $\prod_i (1 + \xi_i t) = \lambda_t(E)$ holds by additivity. The abstract algebra defined by this relation has the same basis by Lemma 4.3, hence is isomorphic to $K^\bullet(X)$.

For a general flag type π , pass from $\underline{D}_\pi(\mathcal{E})$ to the complete flag bundle by successively splitting the $\mathcal{F}_i$. The complete flag K -ring is a free module both over the subalgebra generated by the $\xi_i^{(\alpha)}$ and over $K^\bullet(\underline{D}_\pi(\mathcal{E}))$, with the same basis. Therefore these subrings coincide. The relations follow from Lemma 4.3.

Corollary (4.8). *With the hypotheses above, put*

$$c_i^{(j)} = \gamma^i(F_j - p_j), \quad 1 \leq j \leq k, \quad 1 \leq i \leq p_j.$$

Then $K^\bullet(X)$ is generated over $K^\bullet(S)$ by the $c_i^{(j)}$, and the ideal of relations is generated by those deduced from

$$\gamma_t(E - n) = \prod_{j=1}^k \gamma_t(F_j - p_j). \quad (72)$$

This is just Theorem VI.4.6 rewritten in terms of the γ -operations.

4.9–4.10. Infinite Grassmannians

Let $\underline{G}_n^{r,n}$ be the Grassmannian of type $(n, r - n)$ of $\mathcal{O}_S^r$, with $r \geq n$. As r varies these form an inductive system, denoted $\underline{G}_n^{\infty, n}$, and

$$K^\bullet(\underline{G}_n^{\infty, n}) = \varinjlim_r K^\bullet(\underline{G}_n^{r,n}).$$

The universal quotients give compatible classes c_i in this limit.

Proposition (4.10). *For S quasi-compact there is an isomorphism*

$$K^\bullet(\underline{G}_n^{\infty, n}) \simeq K^\bullet(S)[[c_1, \dots, c_n]]/I,$$

where I is the closed ideal generated by the relations arising from the universal γ -identities for the tautological quotient and its complement.

For finite r the presentation of Theorem VI.4.6 gives

$$K^\bullet(\underline{G}_n^{r,n}) = K^\bullet(S)[c_1, \dots, c_n]/I_r,$$

with I_r homogeneous and supported in weights $> r - n$. Passing to the limit amounts to completing for the topology defined by the I_r . By passing to the splitting algebra and using the nilpotence of the corresponding $\xi_j - 1$, this topology is equivalent to the weight topology, which proves the assertion.

5. Applications to projective bundles; study of f_*

Let S be quasi-compact, E locally free of rank $r + 1$, and $X = \mathbb{P}(E)$ with structural morphism $f : X \rightarrow S$. On X there is the canonical exact sequence

$$0 \rightarrow \mathcal{F} \rightarrow f^*E \rightarrow \mathcal{O}_X(1) \rightarrow 0.$$

Let E, F, ξ denote the corresponding classes in $K^\bullet$. Put $x = \xi - 1$.

5.2

For a locally free sheaf $\mathcal{G}$ on X , the higher direct images $R^i f_* \mathcal{G}$ need not be locally free of finite type on S , so the direct image $f_* : K^\bullet(X) \rightarrow K^\bullet(S)$ is not defined by alternating sums of higher direct images in this setting. Theorem VI.1.1 provides a substitute. Every $y \in K^\bullet(X)$ has a unique expansion

$$y = a_0 + a_1 \xi + \cdots + a_r \xi^r, \quad a_i \in K^\bullet(S), \quad (73)$$

and we put

$$f_*(y) = a_0. \quad (74)$$

This is additive and satisfies the projection formula. Moreover

$$f_*(1) = 1, \quad f_*(\xi^n) = 0 \quad (-r \leq n < 0), \quad f_*(\xi^n) = s_n \quad (n \geq 0), \quad (75)$$

where s_n is the n -th complete symmetric function. The last relations follow by induction from the dependence relation and the Koszul relation (62).

Proposition (5.3). *For every $k \geq 0$,*

$$\mathrm{Fil}^k K^\bullet(X) = \mathrm{Fil}^k K^\bullet(S) + \mathrm{Fil}^{k-1} K^\bullet(S)x + \cdots + \mathrm{Fil}^{k-r} K^\bullet(S)x^r, \quad (76)$$

as a direct sum of $K^\bullet(S)$ -modules, where the filtration is the canonical filtration of augmented λ -algebras.

Here x has zero augmentation and

$$\gamma_t(x) = \lambda_{t/(1-t)}(1 - \xi^{-1}) = \frac{1}{1 - xt},$$

so $\gamma^i(x) = x^i$. The result follows from the filtration computation for free λ -algebras in V, 4.12 and 4.13, together with the relation (64). Directness follows from the independence of $1, \dots, x^r$.

Corollary (5.4). *The filtration induced on $K^\bullet(S)$ by $K^\bullet(X)$ is its original filtration.*

Corollary (5.5). *If X is a product over S of flag bundles of locally free sheaves, then*

$$\mathrm{Gr} K^\bullet(S) \longrightarrow \mathrm{Gr} K^\bullet(X)$$

is injective.

Corollary (5.6). *With the projective-bundle notation above, $\mathrm{Gr} K^\bullet(X)$ is a free $\mathrm{Gr} K^\bullet(S)$ -module with basis $1, x, \dots, x^r$, and*

$$\mathrm{Gr}^k K^\bullet(X) = \mathrm{Gr}^k K^\bullet(S) \oplus \mathrm{Gr}^{k-1} K^\bullet(S)x \oplus \cdots \oplus \mathrm{Gr}^{k-r} K^\bullet(S)x^r.$$

The elements $1, x, \dots, x^{r+1}$ satisfy

$$\sum_{i=0}^{r+1} (-1)^i c_i(E) x^{r+1-i} = 0, \quad (77)$$

where $c_i(E)$ is the i -th Chern class of E .

Corollary (5.7). *For a flag bundle as in 4.5, the $\mathrm{Gr} K^\bullet(S)$ -algebra $\mathrm{Gr} K^\bullet(X)$ is generated by the classes*

$\bar{\xi}_i^{(\alpha)}$ of $\xi_i^{(\alpha)} - 1$, with relations generated by

$$c_t(E) = \prod_{i,\alpha} c_t(\xi_i^{(\alpha)}),$$

where c_t denotes the total Chern class.

Corollary (5.8). *For every k ,*

$$f_*(\mathrm{Fil}^k K^\bullet(X)) \subset \mathrm{Fil}^k K^\bullet(S).$$

This is immediate from Proposition VI.5.3 and the projection formula.

Proposition (5.9). *With the notation of 5.1, for every $y \in K^\bullet(X)$ one has*

$$f_*(\lambda_{-1}(\mathcal{F})y) = 0 \quad \text{and} \quad f_*(\gamma^*(\mathcal{F})y) = 0.$$

In particular,

$$f_*(\lambda_{-1}(\mathcal{F})) = 1.$$

Define coefficientwise

$$\xi_t : 1 + K^\bullet(X)[[t]]^+ \rightarrow 1 + K^\bullet(S)[[t]]^+, \quad \xi_t \left(\sum a_i t^i \right) = \sum f_*(a_i) t^i.$$

Although not generally a group homomorphism, it satisfies

$$\xi_t(m f^* y) = \xi_t(m) y.$$

Using $E = F + \xi$ and $\lambda_t(E) = \lambda_t(F)(1 + \xi t)$ gives the asserted identities.

Proposition (5.10). *Let $z \in K^\bullet(X)$. The condition $z(1 - \xi) = 0$ is equivalent to the existence of $y \in K^\bullet(S)$ such that*

$$(i) \quad z = y \lambda_{-1}(\mathcal{F});$$

$$(ii) \quad y \lambda_{-1}(E) = 0.$$

The sufficiency follows from

$$\lambda_{-1}(E) = \lambda_{-1}(F) \lambda_{-1}(\xi) = \lambda_{-1}(F)(1 - \xi).$$

Conversely write $z = \sum_{i=0}^r a_i \xi^i$. Combining the dependence relation for the ξ^i with $z(1 - \xi) = 0$ and using the independence of $1, \dots, \xi^r$ expresses z as a multiple of $\lambda_{-1}(F)$, and the top relation gives $y \lambda_{-1}(E) = 0$.

Proposition (5.11). *Let $z \in \mathrm{Gr} K^\bullet(X)$. The condition $zx = 0$ is equivalent to the existence of $y \in \mathrm{Gr} K^\bullet(S)$ such that*

$$(i) \quad z = y c^{\mathrm{top}}(F);$$

$$(ii) \quad y c^{\mathrm{top}}(E) = 0.$$

The proof is the same as that of 5.10, after passing to associated graded rings and using 5.6.

6. Study of the filtration of $K^\bullet(X)$ when X has an ample sheaf

Let X be a scheme possessing an ample sheaf $\mathcal{L}$. Then the two definitions of $K^\bullet(X)$, by locally free sheaves of finite type or by perfect complexes, coincide. Let

$$\varepsilon : K^\bullet(X) \rightarrow H^0(X, \mathbb{Z})$$

be the augmentation.

Proposition (6.1). *If X is quasi-compact and has an ample sheaf, then $\text{Fil}^1 K^\bullet(X)$ is a nilideal.*

It suffices to show that $E - \varepsilon(E)$ is nilpotent for the class E of a locally free sheaf. By decomposing X into finitely many disjoint opens and then passing to the flag bundle of E , one reduces to the case where E is the class of an invertible sheaf. Let L be the class of an ample invertible sheaf. For n large, $E^\vee(n)$ is generated by global sections, giving a surjection $(E(-n))^m \rightarrow \mathcal{O}_X$ with locally free kernel. Thus

$$mEL^{-n} = 1 + F.$$

The corresponding γ -relation shows that $1 - EL^{-n}$, hence $L^n - E$, is nilpotent. Since $E - 1 = (L^n - 1) - (L^n - E)$ and $L^n - 1$ is also nilpotent for large n , the result follows.

Proposition (6.2). *Let R be an augmented sub- λ -algebra of $K^\bullet(X)$ generated by the classes of finitely many sheaves. Then the λ -ring filtration of R is discrete and $\text{Fil}^1 R$ is nilpotent. Moreover R is a finite module over its image in $H^0(X, \mathbb{Z})$.*

After reducing to constant ranks, R is generated by finitely many elements $x_i = E_i - \varepsilon(E_i)$ and their γ -operations. Proposition 6.1 makes all these elements nilpotent. Hence the augmentation ideal of R is nilpotent. Since only finitely many non-zero monomials in the x_i and $\gamma^j(x_i)$ remain, the filtration is discrete and R is finite over the augmentation ring.

Corollary (6.3). *Let A be a commutative ring. An element y of $K^\bullet(X) \otimes_{\mathbb{Z}} A$ is invertible if and only if its augmentation is invertible. In particular, the class of any nowhere-zero locally free sheaf of finite type becomes invertible in $K^\bullet(X) \otimes_{\mathbb{Z}} A$.*

The augmentation ideal is nil, so invertibility is detected after quotienting by it.

Corollary (6.4). *Let X be quasi-compact with an ample sheaf, and let $f : X' \rightarrow X$ be proper and perfect. Assume either f is projective or X is noetherian, so that the finiteness theorem applies. If there is $x' \in K^\bullet(X' \otimes A)$ such that the augmentation of $f_*(x')$ is invertible, then*

$$f^* : K^\bullet(X \otimes A) \rightarrow K^\bullet(X' \otimes A)$$

is injective.

Indeed, by the projection formula,

$$f_*(f^*y \cdot x') = y \cdot f_*(x'),$$

and $f_*(x')$ is invertible by 6.3.

6.5–6.6. The codimension filtration

Assume now that X is noetherian of dimension d and has an ample sheaf. Define a filtration $K^\bullet(X)_i$ as follows: $x \in K^\bullet(X)_i$ if and only if for every closed subscheme $Y \subset X$ there is a perfect complex $M^\bullet(Y)$ with

- (a) $\text{cl}(M^\bullet) = x$;
- (b) $\text{codim}(\text{Supp } H^j(M^\bullet) \cap Y, Y) \geq i + j$, with $H^j(M^\bullet) = 0$ for $j \gg 0$.

These $K^\bullet(X)_i$ are additive subgroups.

Proposition (6.6). *The filtration of 6.5 is a ring filtration, and $K^\bullet(X)_{d+1} = 0$.*

If x and x' are represented by perfect complexes $M^\bullet$ and $M'^\bullet$ satisfying the codimension estimates, then xx' is represented by $M^\bullet \otimes M'^\bullet$. The support of $H^n(M^\bullet \otimes M'^\bullet)$ is contained in the union of the intersections of supports with total degree n , and the codimension estimate follows from EGA 0_{IV}, 14.2.2.3. Finally $K^\bullet(X)_{d+1} = 0$ because a support in codimension $> d$ is empty.

Lemma (6.7). *Let $X = \text{Spec } A$ be affine noetherian, let $x_1, \dots, x_n$ be finitely many points of X , and let $\mathcal{E}$ be invertible. There is a section $s \in \Gamma(X, \mathcal{E})$ which is a basis of $\mathcal{E}_{x_i}$ for all i .*

The proof uses closed specializations x'_i with distinct maximal ideals $\mathfrak{m}_i$, the isomorphisms $M/\mathfrak{m}_i M \simeq A/\mathfrak{m}_i$ for $M = \Gamma(X, \mathcal{E})$, and Nakayama's lemma.

Lemma (6.8). *Let X be noetherian with an ample sheaf $\mathcal{L}$, let $\mathcal{E}$ be invertible, and let $x_1, \dots, x_n$ be points of X . There is an integer N such that, for every $k \geq 1$, $\mathcal{E}(kN)$ has a global section which is a basis at every x_i .*

Choose a power of the ample sheaf and a section whose non-vanishing locus is affine and contains all x_i , apply Lemma 6.7 there, and extend after multiplying by a further power.

Theorem (6.9). *Let X be a noetherian scheme of dimension d with an ample sheaf. Then*

$$\text{Fil}^{d+1} K^\bullet(X) = 0,$$

and therefore $(\text{Fil}^1 K^\bullet(X))^{d+1} = 0$.

First, if $\mathcal{E}$ and $\mathcal{E}'$ are invertible, with classes E, E' , then $E - E' \in K^\bullet(X)_1$. For a closed $Y \subset X$, choose one point in each irreducible component. Lemma 6.8 gives morphisms from suitable negative powers of the ample sheaf to $\mathcal{E}$ and $\mathcal{E}'$ which are isomorphisms at those points. The two-term complex built from these morphisms has class $E - E'$ and is acyclic at the chosen generic points of Y , hence its cohomology supports have codimension at least 1 in Y .

Now take $x \in \text{Fil}^{d+1} K^\bullet(X)$. It suffices to treat products

$$\gamma^{i_1}(E_1 - F_1) \cdots \gamma^{i_k}(E_k - F_k), \quad \sum i_j \geq d + 1,$$

with E_i, F_i represented by locally free sheaves of the same constant rank. Pull back to the product of the flag bundles of these sheaves. The pullback is injective on $K^\bullet$ and splits each class into sums of invertible

classes. For invertible classes ξ, η ,

$$\gamma_t(\xi - \eta) = \frac{\gamma_t(\xi - 1)}{\gamma_t(\eta - 1)} = \frac{1 + (\xi - 1)t}{1 + (\eta - 1)t},$$

so $\gamma^i(\xi - \eta) = (-1)^{i-1}(\xi - \eta)(\eta - 1)^{i-1} \in K^\bullet(X')_i$. Thus the pullback of x belongs to $K^\bullet(X')_{d+1}$. Multiplying by a basis element of $K^\bullet(X')$ over $K^\bullet(X)$ of relative codimension r gives an element of $K^\bullet(X')_{d+1+r} = 0$, since $\dim X' = d + r$. By the basis property and injectivity, $x = 0$.

Problem (6.10). *Under the hypotheses of Theorem VI.6.9, is it true that for every k one has*

$$K^\bullet(X)_k = \text{Fil}^k K^\bullet(X)?$$

Bibliography for Expose VI

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SGA 6 – Expose VII

Regular Immersions and the Computation of K' of a Blown-up Scheme

Pierre Berthelot

1. Generalities on regular immersions

The classical notion of regular immersion from EGA IV, 16.9.2, is not adequate in the non-noetherian case. We shall use a weaker notion, which nevertheless has the required properties and which agrees with the usual definition in the noetherian case.

Definition 1.1. Let A be a commutative ring, let E be a finite projective A -module, and let $u : E \rightarrow A$ be an A -linear homomorphism. We say that u is *regular* if the Koszul complex $K_\bullet(u)$ defined by u is acyclic in positive degrees; equivalently, it is a resolution of $B = A/u(E)$.

Recall that the Koszul complex attached to u is the exterior algebra $\wedge(E)$ as a graded A -module, with boundary the interior multiplication by u , considered as an element of $\check{E}$. This is the antiderivation of degree -1 of $\wedge(E)$ which is equal to u on E .

Lemma 1.2. Let $u : E \rightarrow A$ be regular. Let A' be an A -algebra, let $u' : E \otimes_A A' \rightarrow A'$ be obtained from u by base change, and suppose that u' is regular. Put $J = u(E)$ and $J' = JA' = u'(E \otimes_A A')$. Then, for every integer $n \geq 0$, one has

$$\begin{aligned} J^n/J^{n+1} \otimes_A A' &\xrightarrow{\sim} J'^n/J'^{n+1}, \\ \mathrm{Tor}_i^A(J^n/J^{n+1}, A') &= 0 \quad (i \geq 1), \\ J^n \otimes_A A' &\xrightarrow{\sim} J'^n, \\ \mathrm{Tor}_i^A(J^n, A') &= 0 \quad (i \geq 1). \end{aligned}$$

Proof. First suppose that the quotients J^n/J^{n+1} are projective over $B = A/J$. Since J is of finite type, they are locally free of finite type over B . The assertions are local on $\mathrm{Spec} A$, so we may assume that A is local and that $u(E)$ is contained in the maximal ideal. The quotients J^n/J^{n+1} are then free over B , and the vanishing of their higher Tors after tensoring with A' is reduced to the vanishing of $\mathrm{Tor}_i^A(B, A')$. These groups are the homology groups of $K_\bullet(u) \otimes_A A' = K_\bullet(u')$, and vanish in positive degrees by the regularity of u' .

It follows by induction on n , using

$$0 \rightarrow J^n/J^{n+1} \rightarrow A/J^{n+1} \rightarrow A/J^n \rightarrow 0,$$

that $\mathrm{Tor}_i^A(A/J^n, A') = 0$ for $i \geq 1$. The identifications of J^n/J^{n+1} after base change then give the identifications of J^n by using

$$0 \rightarrow J^n \rightarrow A \rightarrow A/J^n \rightarrow 0,$$

and the remaining Tor-vanishing follows from

$$0 \rightarrow J^{n+1} \rightarrow J^n \rightarrow J^n/J^{n+1} \rightarrow 0.$$

It remains to know that J^n/J^{n+1} is projective over B . This is local; assume E is free with basis mapping to $f_1, \dots, f_d$. Let $A_0 = \mathbb{Z}[T_1, \dots, T_d]$ and map T_i to f_i . The corresponding universal homomorphism is regular, and the preceding part applies to the base change $A_0 \rightarrow A$. Since J_0^n/J_0^{n+1} is free of finite type in the universal case, $J^n/J^{n+1} \simeq J_0^n/J_0^{n+1} \otimes_{A_0} A$ is locally free of finite type. This proves the lemma.

Proposition 1.3. With the hypotheses of 1.2, the ideal $J = u(E)$ is quasi-regular in the sense of EGA IV, 16.9.7. Thus J is of finite type, J/J^2 is a finite projective A/J -module, and the canonical homomorphism

$$\mathrm{Sym}_{A/J}(J/J^2) \longrightarrow \mathrm{gr}_J(A) = \bigoplus_{n \geq 0} J^n/J^{n+1}$$

is an isomorphism.

Proof. We have just proved that J/J^2 is finite projective. Locally, using the same universal polynomial algebra as in the proof of 1.2, the displayed morphism is obtained by base change from the corresponding morphism for the ideal generated by the variables. In that case it is an isomorphism. Hence it is an isomorphism in general.

Definition 1.4. Let $(X, \mathcal{A})$ be a commutatively locally ringed site, let $\mathcal{E}$ be an $\mathcal{A}$ -module locally free of finite type, and let $u : \mathcal{E} \rightarrow \mathcal{A}$ be an $\mathcal{A}$ -linear homomorphism. We say that u is regular if its Koszul complex is acyclic in positive degrees. An ideal $\mathcal{J}$ of $\mathcal{A}$ is called regular if it is the image of some surjective regular homomorphism $\mathcal{E} \rightarrow \mathcal{J}$.

Definition 1.5. Let $i : Y \rightarrow X$ be an immersion of schemes. Let U be an open subset of X containing $i(Y)$ such that i is a closed immersion into U . We say that i is *regular* if the

ideal of $\mathcal{O}_U$ defining $i(Y)$ is regular; this condition does not depend on U . The immersion is regular at $y \in Y$ if this is true after restricting to some neighbourhood of $i(y)$.

This definition replaces the one of EGA IV, 16.9.2. In the noetherian case the two definitions are equivalent, by EGA IV, 19.5.1. Any immersion which is locally defined by a regular sequence is regular, and every regular immersion is quasi-regular.

1.4.1–1.4.3. Let $\mathcal{J}$ be a finite type ideal of $\mathcal{O}_X$, and let $x \in \text{Supp}(\mathcal{O}_X/\mathcal{J})$. A family of sections f_i of $\mathcal{J}$ near x is a minimal system of generators near x when its images form a basis of $\mathcal{J} \otimes k(x)$. If $\mathcal{J}/\mathcal{J}^2$ is locally free, a basis of $\mathcal{J}/\mathcal{J}^2$ near x can be lifted to a minimal system of generators of $\mathcal{J}$ near x , by Nakayama's lemma. Thus regularity may be checked on any minimal system of local generators: the corresponding Koszul complex must be acyclic near the point.

Proposition 1.5. A regular immersion remains regular after flat base change. Conversely, regularity may be checked locally for the quasi-compact faithfully flat topology.

Proof. Flat base change carries a Koszul resolution to the base-changed Koszul complex and preserves acyclicity. Conversely, if after a faithfully flat quasi-compact base change the relevant Koszul complex is acyclic, then its positive homology sheaves vanish after a faithful flat pullback and therefore vanish already before base change.

Let $i : Y \rightarrow X$ be a regular closed immersion defined by $\mathcal{J}$. The *conormal sheaf* of Y in X is

$$\underline{N}_{X/Y} = \mathcal{J}/\mathcal{J}^2.$$

It is locally free of finite type. Its rank at a point is the codimension of the immersion at that point. We write it as $\text{codim}(i)$ when it is constant.

Proposition 1.6. Let $i : Y \rightarrow X$ and $j : Z \rightarrow Y$ be two immersions. Then there is a natural exact sequence

$$j^*(\underline{N}_{X/Y}) \longrightarrow \underline{N}_{X/Z} \longrightarrow \underline{N}_{Y/Z} \longrightarrow 0.$$

Proof. Let $\mathcal{I}$ be the ideal of Y in X , $\mathcal{J}$ the ideal of Z in X , and $\mathcal{K}$ the ideal of Z in Y . From

$$0 \rightarrow \mathcal{I} \rightarrow \mathcal{J} \rightarrow \mathcal{K} \rightarrow 0$$

and after tensoring first by $\mathcal{O}_Y$ and then by $\mathcal{O}_Z$, one obtains

$$j^*(\mathcal{I}/\mathcal{I}^2) \rightarrow \mathcal{J}/\mathcal{J}^2 \rightarrow \mathcal{K}/\mathcal{K}^2 \rightarrow 0,$$

which is the asserted sequence.

Proposition 1.7. Let $i : Y \rightarrow X$ and $j : Z \rightarrow Y$ be immersions.

i) If i and j are regular, then $i \circ j$ is regular, and

$$0 \rightarrow j^*(\underline{N}_{X/Y}) \rightarrow \underline{N}_{X/Z} \rightarrow \underline{N}_{Y/Z} \rightarrow 0$$

is exact. In particular $\text{codim}(i \circ j) = \text{codim}(i) + \text{codim}(j)$.

ii) If $i \circ j$ and i are regular, and the preceding conormal sequence is exact and locally split, then j is regular.

iii) If X is noetherian, and if $i \circ j$ and j are regular, then i is regular at the points of Z .

Proof. The assertion is local, so write $X = \text{Spec } A$, $Y = \text{Spec } B$, $Z = \text{Spec } C$, with $B = A/I$ and $C = A/J = B/K$. Choose minimal systems of generators $f_1, \dots, f_n$ for I and $\bar{g}_1, \dots, \bar{g}_m$ for K , lifted to elements g_j of A . The situation is obtained from the standard diagram

$$\mathbb{Z}[T_1, \dots, T_{n+m}] \rightarrow \mathbb{Z}[T_1, \dots, T_m] \rightarrow \mathbb{Z}$$

by base change. The regularity conditions are expressed by the vanishing of the corresponding Tor groups. The transitivity spectral sequence

$$E_{pq}^2 = \text{Tor}_p^{B_0}(C_0, \text{Tor}_q^{A_0}(B_0, A)) \implies \text{Tor}_{p+q}^{A_0}(C_0, A)$$

then gives (i) and (ii). The rank computation for the conormal sheaves gives exactness on the left. Part (iii) is the noetherian statement of EGA IV, 19.1.5, using the equivalence of definitions in the noetherian case.

Proposition 1.8. Let X be a scheme, let $\mathcal{E}$ be a locally free $\mathcal{O}_X$ -module of finite type, and let $u : \mathcal{E} \rightarrow \mathcal{O}_X$ be regular with image $\mathcal{J}$. Let Y be the closed subscheme defined by $\mathcal{J}$, let X' be the blow-up of X along Y , let $P = \mathbb{P}(\mathcal{E})$ with structural map $p : P \rightarrow X$, and consider on P the exact sequence

$$0 \rightarrow \mathcal{F} \rightarrow p^*\mathcal{E} \xrightarrow{v} \mathcal{O}_P(1) \rightarrow 0.$$

Then:

i) The formation of X' commutes with every base change $T \rightarrow X$ for which the pullback of u is regular.

- ii) The closed immersion $j : X' \rightarrow P$ defined by the surjection $S(\mathcal{E}) \rightarrow \bigoplus_{n \geq 0} \mathcal{J}^n$ is regular. Its ideal is the image of the regular morphism v ; if $\text{codim}(Y, X)$ is constant, then $\text{codim}(j) = \text{codim}(i) - 1$.
- iii) The graded $S(\mathcal{E})$ -module $\bigoplus_{n \geq 0} \mathcal{J}^n$ is perfect as a graded module: locally it has a finite graded resolution by graded free modules of finite type.

Proof. The first assertion follows from $X' = \text{Proj}(\bigoplus \mathcal{J}^n)$ and the base-change formulas of 1.2. For (ii), the question is local. In the standard case $A = \mathbb{Z}[X_1, \dots, X_d]$, with $u(e_i) = X_i$, one works on the standard open $D^+(X_1)$ of the projective bundle. The kernel of $p^*\mathcal{E} \rightarrow \mathcal{O}_P(1)$ is generated by $e_i - (X_i/X_1)e_1$, $i > 1$, and the functions defining the image of v form a regular sequence; they generate exactly the ideal of the blow-up in the projective bundle. The general case is obtained by base change from this standard case, and acyclicity follows from the Tor-vanishing in 1.2. For (iii), construct in the standard noetherian case a finite graded free resolution, and tensor it with the given ring. The homology in positive degree is computed by the Tor groups $\text{Tor}_i(A, \mathcal{J}_0^n)$, which vanish by 1.2; hence the tensor complex is the desired resolution.

Proposition 1.9. Let $i : Y \rightarrow X$ be a regular closed immersion, let X' be the blow-up of X along Y , and let $f : X' \rightarrow X$ be the structural morphism. Then i and f are perfect morphisms.

Proof. Locally the structure sheaf of Y has a finite locally free resolution given by a Koszul complex, hence i is perfect. By 1.8, f factors as a regular immersion into a projective bundle followed by the smooth structural morphism of the projective bundle. Both factors are perfect.

Proposition 1.10. If X and Y are smooth schemes over S , then every immersion $X \rightarrow Y$ is regular. In particular every section of a smooth morphism is a regular immersion.

This follows at once from EGA IV, 17.12.1, because in the noetherian smooth case the EGA notion of regular immersion agrees with the present one.

2. Computations for regular immersions

2.1–2.3. Let A be a ring and let B, C be A -algebras. The graded module

$$\bigoplus_{i \geq 0} \text{Tor}_i^A(B, C)$$

has a natural structure of an associative graded algebra over $B \otimes_A C$. Indeed, take a projective resolution $P_\bullet$ of B . The product is obtained by composing the canonical homology product

$$H_i(P_\bullet \otimes_A C) \otimes_A H_j(P_\bullet \otimes_A C) \rightarrow H_{i+j}((P_\bullet \otimes_A C) \otimes_A (P_\bullet \otimes_A C))$$

with a lifting $P_\bullet \otimes_A P_\bullet \rightarrow P_\bullet$ of the multiplication $B \otimes_A B \rightarrow B$, and then with the multiplication of C . The result is independent of choices and is functorial in B and C .

If the resolution $P_\bullet$ is itself a differential graded algebra, then this product is computed directly by the product of $P_\bullet$. In particular, for a quotient by a regular ideal one may use the Koszul resolution; hence the resulting Tor-algebra is alternating. These constructions globalize: for two X -schemes X' and X'' , the sheaf

$$\bigoplus_{i \geq 0} \mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_{X''})$$

is a graded associative $\mathcal{O}_{X' \times_X X''}$ -algebra.

Lemma 2.4. Let X' and X'' be two X -schemes, and let $\mathcal{M}$ be a flat $\mathcal{O}_{X'}$ -module. Then there is a canonical isomorphism

$$\mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_{X''}) \otimes_{\mathcal{O}_{X'}} \mathcal{M} \xrightarrow{\sim} \mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{M}, \mathcal{O}_{X''}).$$

Proof. This is affine-local. If $X = \mathrm{Spec} A$, $X' = \mathrm{Spec} A'$, $X'' = \mathrm{Spec} A''$, and M is a flat A' -module, a projective A -resolution of A'' gives

$$M \otimes_{A'} (A' \otimes_A P_\bullet) \simeq M \otimes_A P_\bullet,$$

and passage to homology gives the assertion.

Proposition 2.5. Let Y and Z be closed subschemes of X , and put $T = Y \times_X Z$. Assume that locally near every point of T the schemes Y , Z , and T are defined by Koszul resolutions associated to systems of sections $f_1, \dots, f_n$, $g_1, \dots, g_m$, and their union. Then

$$\mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_Y, \mathcal{O}_Z) \simeq \bigwedge^i \mathrm{Tor}_1^{\mathcal{O}_X}(\mathcal{O}_Y, \mathcal{O}_Z)$$

for all $i \geq 0$, and

$$\mathrm{Tor}_1^{\mathcal{O}_X}(\mathcal{O}_Y, \mathcal{O}_Z) \simeq \mathrm{Ker}(i^* \underline{N}_{X/Y} \oplus j^* \underline{N}_{X/Z} \rightarrow \underline{N}_{X/T}),$$

where $i : T \rightarrow Y$ and $j : T \rightarrow Z$ are the induced immersions. In particular, if $Y \rightarrow X$ is a regular closed immersion, then

$$\mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_Y, \mathcal{O}_Z) \simeq \bigwedge^i \mathrm{Tor}_1^{\mathcal{O}_X}(\mathcal{O}_Y, \mathcal{O}_Z), \quad \mathrm{Tor}_1^{\mathcal{O}_X}(\mathcal{O}_Y, \mathcal{O}_Z) \simeq i^* \underline{N}_{X/Y}.$$

Proof. The Tor-algebra is alternating by the Koszul calculation of 2.2, hence there is a unique algebra homomorphism from the exterior algebra on degree one. Locally, the tensor product of the two Koszul complexes is the Koszul complex for the union of the two systems. The spectral sequence of the associated bicomplex degenerates, and the homology is exactly the exterior algebra on the degree-one homology. The description of degree one follows from the exact sequence

$$0 \rightarrow \mathcal{I} \cap \mathcal{J} \rightarrow \mathcal{I} \oplus \mathcal{J} \rightarrow \mathcal{I} + \mathcal{J} \rightarrow 0$$

and the Tor exact sequence; it identifies the degree-one Tor with the kernel of the difference map between the two conormal sheaves and the conormal sheaf of T .

2.6. Suppose that X has an ample sheaf. If $i : Y \rightarrow X$ is a regular closed immersion, then i is perfect and the additive morphism

$$i_* : K'(Y) \rightarrow K'(X)$$

is defined by sending a perfect complex on Y to its direct image on X .

Proposition 2.7. Let X have an ample sheaf, let $i : Y \rightarrow X$ be a regular closed immersion, and let $\mathcal{N}$ be the conormal sheaf of Y in X . Then, for every $x \in K^\bullet(Y)$,

$$i^! i_*(x) = x \cdot \lambda_{-1}(\mathcal{N}), \quad \lambda_{-1}(\mathcal{N}) = \sum_{i \geq 0} (-1)^i \lambda^i(\mathcal{N}).$$

Proof. By additivity it is enough to take x to be the class of a locally free $\mathcal{O}_Y$ -module $\mathcal{M}$ of finite type. The homology sheaves of $\mathrm{Li}^* \mathrm{R}i_*(\mathcal{M})$ are

$$\mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{M}, \mathcal{O}_Y) \simeq \mathcal{M} \otimes_{\mathcal{O}_Y} \bigwedge^i \mathcal{N}$$

by 2.4 and 2.5. Their alternating sum in $K'(Y)$ is precisely $x \lambda_{-1}(\mathcal{N})$.

Corollary 2.8. Under the hypotheses of 2.7, the self-intersection formula is

$$i_*(y \cdot x \cdot \lambda_{-1}(\mathcal{N})) = i_*(y) \cdot i_*(x)$$

for $x, y \in K^\bullet(Y)$. This follows from 2.7 and the projection formula.

3. Computation of K' of a blow-up

Let $i : Y \rightarrow X$ be a regular closed immersion. Let X' be the blow-up of X along Y , put $Y' = Y \times_X X'$, and write $f : X' \rightarrow X$ and $g : Y' \rightarrow Y$ for the structural morphisms. The immersion $j : Y' \rightarrow X'$ is a regular closed immersion of codimension one, defined by an invertible ideal. If $\mathcal{J}$ is the ideal of Y in X , then $\mathcal{J}\mathcal{O}_{X'} = \mathcal{O}_{X'}(1)$.

Since

$$Y' = \text{Proj}\left(\bigoplus_{n \geq 0} \mathcal{J}^n / \mathcal{J}^{n+1}\right) = \text{Proj}(\text{Sym}(\mathcal{N})) = \mathbb{P}(\mathcal{N}),$$

where $\mathcal{N} = \mathcal{J}/\mathcal{J}^2$, there is on Y' an exact sequence

$$0 \rightarrow \mathcal{F} \rightarrow g^*\mathcal{N} \rightarrow \mathcal{O}_{Y'}(1) \rightarrow 0.$$

The sheaf $\mathcal{O}_{Y'}(1)$ is also the conormal sheaf of j .

Lemma 3.2. With these notations,

$$\text{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_Y) \simeq \bigwedge^i \text{Tor}_1^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_Y), \quad \text{Tor}_1^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_Y) \simeq \mathcal{F}.$$

Proof. Work locally where $\mathcal{J}$ is defined by a regular sequence $f_1, \dots, f_d$. On the standard open of the blow-up on which $f_1^{(1)}$ is invertible, change the basis of the free module defining the Koszul complex by setting $e'_1 = e_1$ and $e'_i = e_i - (f_i/f_1)e_1$ for $i > 1$. The differential sends e'_1 to f_1 and the other e'_i to zero. Since f_1 is a non-zero-divisor on that open, the homology is the exterior algebra on the image of the module spanned by $e'_2, \dots, e'_d$, which is $\mathcal{F}$. These local identifications are compatible with the algebra structure and glue.

3.3. If X has an ample sheaf, then so do X' , Y , and Y' . The immersions i and j , the morphism g (a projective bundle), and the morphism f are perfect. Hence we have additive maps

$$i_* : K'(Y) \rightarrow K'(X), \quad j_* : K'(Y') \rightarrow K'(X'), \quad f_* : K'(X') \rightarrow K'(X), \quad g_* : K'(Y') \rightarrow K'(Y).$$

Proposition 3.4. For every $x \in K'(Y)$,

$$f^*i_*(x) = j_*(g^*(x)\lambda_{-1}(\mathcal{F})).$$

Equivalently, after applying f_* to the two sides, this is the usual compatibility of the Gysin construction with the exceptional divisor.

Proof. It is enough by additivity to take x to be a locally free sheaf $\mathcal{M}$ on Y . The homology of $Lf^*Ri_*(\mathcal{M})$ is $\mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{M}, \mathcal{O}_{X'})$, and 2.4 together with Lemma 3.2 identifies this with $g^*\mathcal{M} \otimes \wedge^i \mathcal{F}$. Taking the alternating class gives the displayed formula.

Lemma 3.5. For the blow-up $f : X' \rightarrow X$,

$$\text{i) } R^i f_* \mathcal{O}_{X'}(n) = 0 \text{ for } i \geq 1 \text{ and } n \geq 0;$$

$$\text{ii) } f_* \mathcal{O}_{X'}(n) = \mathcal{J}^n \text{ for } n \geq 0.$$

Proof. The assertion is local on X . For $n \gg 0$ the vanishing follows from the standard acyclicity theorem for projective perfect morphisms. The exact sequences

$$0 \rightarrow \mathcal{O}_{X'}(n+1) \rightarrow \mathcal{O}_{X'}(n) \rightarrow \mathcal{O}_{Y'}(n) \rightarrow 0$$

and the known cohomology of $\mathcal{O}(n)$ on the projective bundle $Y' = \mathbb{P}(\mathcal{N})$ permit descending induction to all $n \geq 0$. The identification of the direct image with $\mathcal{J}^n$ is first obtained for large n from a finite graded presentation of $\bigoplus \mathcal{J}^n$, then descended by the same exact sequences.

Proposition 3.6. With the above notation,

$$f_* f^* = \mathrm{Id}_{K'(X)}.$$

Proof. The projection formula gives $f_* f^*(x) = x \cdot f_*(1)$. By Lemma 3.5, $Rf_* \mathcal{O}_{X'}$ has no higher cohomology and degree-zero cohomology $\mathcal{O}_X$, so $f_*(1) = 1$.

Theorem 3.7. Suppose X is quasi-compact and has an ample sheaf. Then the sequence

$$0 \longrightarrow K'(Y) \xrightarrow{u} K'(Y') \times K'(X) \xrightarrow{v} K'(X')$$

is exact, where

$$u(y) = (g^*(y)\lambda_{-1}(\mathcal{F}), -i_*(y)), \quad v(y', x) = j_*(y') + f^*(x).$$

Moreover u admits the left inverse $u'(y', x) = g_*(y')$. If, in addition, X is noetherian and Y is regular, then v is surjective, and hence

$$0 \rightarrow K'(Y) \rightarrow K'(Y') \times K'(X) \rightarrow K'(X') \rightarrow 0$$

is exact.

Proof. The identity $g_*\lambda_{-1}(\mathcal{F}) = 1$, together with the projection formula, gives the left inverse of u . The equality $v \circ u = 0$ is exactly Proposition 3.4. Conversely, if $j_*(y') + f^*(x) = 0$, applying f_* and using 3.6 gives $x = -i_*g_*(y')$. Put $y = g_*(y')$. It remains to prove $y' = g^*(y)\lambda_{-1}(\mathcal{F})$. The difference $z = y' - g^*(y)\lambda_{-1}(\mathcal{F})$ has $g_*(z) = 0$ and $j_*(z) = 0$. The self-intersection formula gives $j^*j_*(z) = z(1 - L)$, where L is the conormal line of j . The projective-bundle computation implies that the simultaneous kernel of j_* and g_* is zero; hence $z = 0$.

If X is noetherian and Y regular, then Y' and X' are regular along the exceptional divisor, and the usual devissage and localization argument gives the surjectivity of v .

Remark 3.8. In the noetherian regular case, Theorem 3.7 determines the abelian group $K'(X')$ from the λ -rings $K'(X)$ and $K'(Y)$, the maps i_* and i^* , and the class of the conormal sheaf N in $K'(Y)$. The projective-bundle formula computes $K'(Y')$ from $K'(Y)$ and N . The next results describe the multiplicative and rational λ -structures from the same data.

Proposition 3.9. Let X be noetherian and Y regular. In the notation of 3.7, the exterior-power operations on the class represented by $f^*(x) = j_*(y') + x'$ are computed by the same universal formulas as in the projective-bundle case: if $y = g_*(y')$, then

$$f^*(\lambda^p(x)) = j_* \left(\sum_{i=0}^{p-1} (-1)^i \lambda^{p-1-i}(N) g^*(\lambda^i(y)) \right) + \lambda^p(x')$$

whenever the displayed decomposition is defined. The proof reduces to the case where i is a section of a projective bundle, by using the blow-up square and functoriality of the operations λ^p .

4. Self-intersection formulas for the operations λ^p and γ^p

Proposition 4.1. Let $i : Y \rightarrow X$ be a regular immersion of codimension d , and suppose the ideal of Y is generated by a regular sequence of d sections. If N is the conormal class

and $y \in K'(Y)$, then for every $p \geq 1$,

$$i^*i_*(\lambda^p(Ny)) = \lambda^p(N)\lambda^p(y).$$

The proof reduces to the zero section of a projective space bundle.

Propositions 4.2–4.3. The preceding formula holds for every regular immersion. One embeds the situation into successive projective bundles so that the composite immersion is given by a regular sequence, proves the formula for the composite, and descends step by step.

Consequently, for every regular immersion $i : Y \rightarrow X$ and every $y \in K'(Y)$,

$$i_*(\lambda^p(Ny)) = \sum_{k=0}^p (-1)^k \lambda^{p-k}(N) \lambda^k(i_*(y)) \quad \text{modulo torsion.}$$

Theorem 4.3. Let $i : Y \rightarrow X$ be a regular immersion with conormal class N . For every $y \in K'(Y)$ and every $p \geq 1$,

$$i_*(\gamma^p(Ny)) = \gamma^p(i_*(y)) \quad \text{modulo torsion.}$$

Proof. Apply the previous exterior-power formula to the generating functions defining the Adams and Grothendieck operations. If

$$u(t) = 1 + \sum_{k \geq 1} i_*(\lambda^k(Ny))t^k, \quad v(t) = \sum_{k \geq 0} \lambda^k(i_*(y))t^k,$$

then the logarithmic-derivative relation implies $d(u(t)/v(t))/dt \equiv 0$ modulo torsion. Since the constant term is 1, integration gives $u(t) \equiv v(t)$, which is equivalent to the stated congruences for the γ^p .

Corollary 4.4. The same congruence follows directly for the operations γ^p , since they are integral linear combinations of the exterior-power operations in the universal formulas.

Remark 4.5. The proof suggests that the congruence should in fact be an equality, not merely an equality modulo torsion. In characteristic zero this was proved in Riemann-Roch references by a geometric characterization of the classes $\lambda^p(Nx)$. Such a method should extend to all characteristics once exterior powers are available for perfect complexes.

4.6. Application to the multiplicative structure

Keep the notation of 3.1 and 3.3, and suppose X noetherian and Y regular. Theorem 3.7 describes $K'(X')$ as an abelian group. Its multiplication is determined by the rings $K'(X)$ and $K'(Y)$, the maps i_* and i^* , and the conormal class N .

More explicitly, for $y'_1, y'_2 \in K'(Y')$ and $x_1, x_2 \in K'(X)$,

$$\begin{aligned} & (j_*(y'_1) + f^*(x_1))(j_*(y'_2) + f^*(x_2)) \\ &= j_*(y'_1 g^*(g_*(y'_2)) \lambda_{-1}(\mathcal{F})) + j_*(y'_1 f^* i^*(x_2)) + j_*(y'_2 f^* i^*(x_1)) + f^*(x_1 x_2). \end{aligned}$$

This is just the projection formula, the self-intersection formula for j , and Theorem 3.7.

4.7. Rational λ -ring structure of $K'(X')$

After tensoring with $\mathbb{Q}$, the λ -ring structure of $K'(X')$ is determined by the same data. Recall that over a $\mathbb{Q}$ -algebra the datum of a binomial-ring structure is equivalent to a λ -ring structure. If $y' \in K'(Y')$ and $y = g_*(y')$, then

$$j^* j_*(y') = y' g^*(N)$$

and the Adams operations satisfy

$$\psi^k(j_*(y')) = j_*(g^*(\psi^k(y)) \lambda_{-1}(\mathcal{F})^{k-1}).$$

Together with the formulas of 3.9 and the known λ -structures on $K'(X)_{\mathbb{Q}}$ and $K'(Y)_{\mathbb{Q}}$, these identities determine the operations λ^p on $K'(X')_{\mathbb{Q}}$.

5. Complements and applications

5.1. Complete intersections

Proposition 5.1. Let X be a regular scheme with an ample sheaf, and let Y be a closed subscheme which is a complete intersection of order r . Then the ideal of Y is locally generated by an $\mathcal{O}_X$ -regular sequence of r sections.

Theorem 5.2. Let X be a regular scheme with an ample sheaf, let Z be an irreducible closed subscheme of X , and let z be its generic point. Suppose the maximal ideal of $\mathcal{O}_{X,z}$

can be generated by an $\mathcal{O}_{X,z}$ -regular sequence. Then there is an open neighbourhood U of z such that $Z \cap U \rightarrow U$ is a regular immersion.

Proof. The ideal of Z at its generic point is the maximal ideal of the regular local ring $\mathcal{O}_{X,z}$, and by hypothesis it has a regular system of generators. After shrinking X near z , these generators extend to sections defining a regular ideal. Hence $Z \cap U \rightarrow U$ is regular, of the expected codimension.

5.3. Filtrations and Gysin morphisms

For the filtration by codimension of support, a regular immersion $i : Y \rightarrow X$ of codimension d satisfies

$$i_*(\mathrm{Fil}^k K'(Y)) \subset \mathrm{Fil}^{k+d} K'(X).$$

For the blow-up $f : X' \rightarrow X$ along Y ,

$$f^*(\mathrm{Fil}^k K'(X)) \subset \mathrm{Fil}^k K'(X').$$

The exact sequence of Theorem 3.7 is compatible with these filtrations in the noetherian regular case.

5.6. Surjectivity after tensoring with $\mathbb{Q}$

Suppose X is noetherian and Y regular. The surjectivity in Theorem 3.7 remains true after tensoring with $\mathbb{Q}$. If $x' \in \mathrm{Fil}^k K'(X')_{\mathbb{Q}}$, one may choose $(y', x) \in K'(Y')_{\mathbb{Q}} \times K'(X)_{\mathbb{Q}}$ with

$$x' = v(y', x), \quad x \in \mathrm{Fil}^k K'(X)_{\mathbb{Q}}.$$

Filtering the product by

$$\mathrm{Fil}^k(K'(Y')_{\mathbb{Q}} \times K'(X)_{\mathbb{Q}}) = \mathrm{Fil}^{k-1} K'(Y')_{\mathbb{Q}} \times \mathrm{Fil}^k K'(X)_{\mathbb{Q}},$$

one obtains a surjection

$$\mathrm{Fil}^k(K'(Y')_{\mathbb{Q}} \times K'(X)_{\mathbb{Q}}) \longrightarrow \mathrm{Fil}^k K'(X')_{\mathbb{Q}}.$$

Indeed, if $x' = \gamma^k(x'_1)$ with augmentation zero, lift x'_1 to a pair (y'_1, x_1) with x_1 of filtration at least one; the formula for the γ -operation in the product ring gives a lift of x' in the required filtered degree.

SGA 6 – Expose VIII

The Riemann–Roch Theorem

1. Complete-intersection morphisms

Definition 1.1. Let $f : X \rightarrow Y$ be a morphism of schemes. We say that f is a *complete-intersection morphism* if, for every point $x \in X$, there exist an open neighbourhood U of x and a smooth Y -scheme V such that the restriction $f|_U$ factors as

$$U \xrightarrow{i} V \rightarrow Y,$$

where i is a regular immersion in the sense of Expose VII. After replacing V by an open subscheme, one may suppose that i is a regular closed immersion.

In this definition one may replace V by an affine space $\mathbb{A}_Y^n$. Indeed, locally V embeds as a closed subscheme of an affine space over an affine open of Y ; this immersion is regular because V is smooth, and the composite of regular immersions is regular.

Proposition 1.2. Let $f : X \rightarrow Y$ be a morphism. Suppose that two factorizations of f through smooth Y -schemes are related by a smooth morphism between the intermediate schemes. Then the immersion in one factorization is regular if and only if the immersion in the other factorization is regular.

Proof. If $i : X \rightarrow V$ is regular and $V \rightarrow V'$ is smooth, form $X' = V \times_{V'} X$. The induced section $X \rightarrow X'$ is a regular immersion, being a section of a smooth morphism, and the base change of i is regular. Hence the composite immersion is regular. Conversely, when the smooth morphism is etale the comparison is immediate because the induced section is open; the general smooth case is reduced etale-locally to this situation.

Proposition 1.3. Suppose $X = \operatorname{Spec} B$, $Y = \operatorname{Spec} A$, and $f : X \rightarrow Y$ factors through a regular closed immersion $j : X \rightarrow \mathbb{A}_Y^n$. Then one may find an analogous factorization through $\mathbb{A}_Y^{n+1}$ in which a section of the projection is a regular immersion and the remaining immersion is regular.

Proof. Choose lifts in A of the coordinate functions defining the original factorization. They define a section of the larger affine space. Since a section of a smooth morphism is a regular immersion, and since the conormal sequence for the two immersions is exact and locally split, the remaining immersion is regular by Expose VII, 1.7.

Proposition 1.4. The composite of two complete-intersection morphisms is a complete-intersection morphism.

Proof. Factor each morphism locally as a regular immersion followed by a smooth morphism. After restricting to suitable opens, the composite factors through the fiber product of the two smooth intermediates; the new intermediate is smooth and the induced immersion is regular by Expose VII, 1.7.

Proposition 1.5. Complete-intersection morphisms are stable under arbitrary base change.

Proof. This follows from the definition, from the stability of smooth morphisms under base change, and from the flat-base-change stability of regular immersions.

Proposition 1.6. A morphism which locally factors as a regular immersion followed by a smooth morphism is perfect. In particular every complete-intersection morphism is perfect.

Proof. A regular immersion is perfect by Expose VII, 1.9, and a smooth morphism is perfect. The composite of perfect morphisms is perfect.

Proposition 1.7. If a complete-intersection morphism is flat and of finite presentation, then locally it is a regular immersion followed by a smooth morphism of fixed pure relative dimension.

Proposition 1.8. Let $f : X \rightarrow Y$ be a complete-intersection morphism, and let

$$U \xrightarrow{i} V \xrightarrow{g} Y$$

be a local factorization near x , with i regular and g smooth. If $\mathcal{J}$ is the ideal of i , the integer

$$d_x = \text{rank}(\Omega_{V/Y}^1)_{i(x)} - \text{rank}(\mathcal{J}/\mathcal{J}^2)_x$$

does not depend on the chosen factorization.

Proof. Given two factorizations, compare them through the fiber product of the two smooth intermediate schemes over Y . The conormal sheaf for the induced immersion is obtained by adding the pullback of the relative differentials of the smooth comparison morphism. Hence the displayed difference of ranks is unchanged.

Definition 1.9. The integer d_x is called the *virtual relative dimension* of f at x . If it is independent of x , for instance when X is connected, it is called the virtual relative dimension

of f and is denoted $d(f)$ or $d_{X/Y}$.

Proposition 1.10. If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are complete-intersection morphisms, then

$$d_{g \circ f} = d_f + d_g.$$

Proposition 1.11. In a cartesian square

$$\begin{array}{ccc} X' & \longrightarrow & X \\ \downarrow & & \downarrow f \\ Y' & \longrightarrow & Y, \end{array}$$

if f is a complete-intersection morphism, then so is its base change f' , and $d_{f'} = d_f$.

2. Projective morphisms and complete-intersection morphisms

Proposition 2.1. A projective smooth morphism is a complete-intersection morphism, and its virtual relative dimension is its relative dimension.

Proposition 2.2. A projective morphism which is a regular closed immersion is a complete-intersection morphism of virtual relative dimension $-\text{codim}(X, Y)$.

Theorem 2.3. Let $f : X \rightarrow Y$ be projective. Suppose f factors as

$$X \xrightarrow{i} \mathbb{P}(E) \xrightarrow{p} Y,$$

where E is a vector bundle on Y , p is the canonical projection, and i is a regular immersion. Then f is a complete-intersection morphism and

$$d_f = \text{rank}(E) - 1 - \text{codim}(X, \mathbb{P}(E)).$$

Proof. The projection p is smooth of relative dimension $\text{rank}(E) - 1$, and the immersion i has virtual relative dimension equal to the negative of its codimension. The formula follows from the additivity of virtual dimensions.

Corollary 2.4. If $f : X \rightarrow Y$ is projective and a complete-intersection morphism, then after choosing a sufficiently ample vector bundle on Y , f may be factored as a regular immersion into a projective bundle followed by the projection.

Proposition 2.5. Let $f : X \rightarrow Y$ be projective and a complete-intersection morphism.

- i) If f is finite, then $d_f \leq 0$.
- ii) If f is surjective, then $d_f \geq 0$.
- iii) If f is finite and surjective, then $d_f = 0$.

Remark 2.6. If Y is connected and $f : X \rightarrow Y$ is projective and a complete-intersection morphism, then d_f is constant on Y ; for a dominant morphism it is $\dim X - \dim Y$.

3. Complete-intersection morphisms and the Chern character

Proposition 3.1. Let $f : X \rightarrow Y$ be a projective complete-intersection morphism of virtual relative dimension d . Suppose Y is quasi-compact and has an ample sheaf. Then, for every integer k ,

$$f_*(\mathrm{Fil}^k K'(X)_{\mathbb{Q}}) \subset \mathrm{Fil}^{k-d} K'(Y)_{\mathbb{Q}}.$$

Proof. Factor f as $X \xrightarrow{i} P \xrightarrow{p} Y$, with p smooth of relative dimension n and i a regular immersion of codimension $n - d$. The Gysin map for i raises the support-codimension filtration by $n - d$, while the smooth projective projection p lowers it by n . The result follows.

Proposition 3.2. The preceding inclusion defines graded homomorphisms

$$f_{\mathrm{gr}} : \mathrm{Gr}^k K'(X)_{\mathbb{Q}} \longrightarrow \mathrm{Gr}^{k-d} K'(Y)_{\mathbb{Q}}.$$

3.3. Reminder on the Chern character and the Todd class

For a graded K -algebra A , with K a binomial ring, Expose V defined

$$\mathrm{Ch}(A) = K \times (1 + \hat{A}^+).$$

For every augmented K - λ -algebra Λ there is a K - λ -homomorphism, the completed Chern class,

$$\tilde{c} = (\varepsilon, c) : \Lambda \longrightarrow \text{Ch}(\text{Gr}_\Lambda).$$

There is also a functorial ring homomorphism, the Chern character,

$$\text{ch} : \text{Ch}(A) \longrightarrow K \otimes_{\mathbb{Z}} \mathbb{Q},$$

and a multiplicative endomorphism, the Todd character,

$$\text{Td} : 1 + \hat{A}^+ \longrightarrow 1 + \hat{A}^+.$$

Both are multiplicative in the expected sense: $\text{ch}(xy) = \text{ch}(x)\text{ch}(y)$ and $\text{Td}(x \oplus y) = \text{Td}(x)\text{Td}(y)$.

3.4. The Todd character of a morphism

Definition 3.4. Let $f : X \rightarrow Y$ be a complete-intersection morphism of virtual relative dimension d . Assume Y is quasi-compact and has an ample sheaf. The *Todd character* of f , denoted $\tau(f)$ or $\tau_{X/Y}$, is the element of $\text{Gr } K'(X)_{\mathbb{Q}}$ defined by

$$\tau(f) = \text{Td}(N_f)^{-1},$$

where N_f is the virtual conormal class of f in $K'(X)$.

Proposition 3.4. If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are complete-intersection morphisms, then

$$\tau(g \circ f) = \tau(f) \cdot f^* \tau(g).$$

Proof. The virtual conormal class satisfies $N_{g \circ f} = N_f \oplus f^* N_g$. The multiplicativity of the Todd character gives

$$\text{Td}(N_{g \circ f}) = \text{Td}(N_f) f^* \text{Td}(N_g),$$

and taking inverses gives the formula.